

Designing a potentiostat for differential pulse voltammetry

by

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> *A very long quote*
> *from someone very important.*
> – Mr M

Thank you to Leanne for helping to prepare this template.

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Abstract

Uittreksel

Table of Contents

Abstract **iii**

Uittreksel **iv**

Table of Contents **v**

List of Figures **x**

List of Tables **xi**

Nomenclature **xiv**

Acronyms and Abbreviations xv

Definitions xvii

1. Introduction **1**

1.1. Background and Motivation 1

1.2. Problem Statement 1

1.3. Project Aim, Objectives and Scope 2

1.4. Scope Exclusions 2

2. Literature **3**

2.1. Electrochemical Sensing 3

2.1.1. Voltammetry as a Family of Techniques 4

2.1.2. Equivalent Circuit of the Cell 4

2.1.3. Differential Pulse Voltammetry 5

2.1.4. Application to Antiviral Drug Residues in Water 6

2.2. Potentiostat Fundamentals 7
2.2.1. The Role of a Potentiostat 7
2.2.2. Three Operational Amplifier Potentiostat Topology 8
2.2.3. Key Performance Parameters 8
2.2.4. Analog Front-End Approaches 9
2.3. Survey of Existing Potentiostat Designs 9
2.3.1. Commercial Instruments 10
2.3.2. Discrete Open-Source Designs 10
2.3.3. AFE-Based and Web-Hosted Designs 11

3. System Modeling & Design **12**
3.1. Signal Chain 12
3.2. Choice and Operating Limits of the TIA 13
3.2.1. High-Speed or Low-Power TIA 13
3.2.2. Stability with a Cell Connected 13
3.2.3. Analog and Digital Bandwidth 15
3.3. Summary 15

4. Hardware Design **16**
4.1. AFE Requirements 16
4.2. System Requirements 17
4.3. Microcontroller 17
4.4. PCB Component Placement and Layout 18
4.4.1. Capacitor Choices 18
4.4.2. Ferrite Bead 19
4.4.3. Analog and Digital Ground 20
4.4.4. Trace Widths 20

4.5. Display and SD Card 21
4.6. Power Management 21
4.6.1. Battery 21
4.6.2. Charging 22
4.6.3. Regulation 22
4.6.4. Input Decoupling 23
4.7. User Interface 23

5. Software Design **24**
5.1. Software Overview 24
5.2. Top-Level Program Flow 25
5.3. DPV Measurement Sequence 25
5.3.1. Generating the Potential 26
5.3.2. Scan Flow 27
5.4. Pulse Sampling and Timing 27
5.4.1. Software Timing 27
5.4.2. Averaging Window 28
5.4.3. Placing Windows 28
5.4.4. Limitations 29
5.5. Current Measurement and Ranging 30
5.5.1. Ranges and Calibration 30
5.5.2. Automatic Range Selection 30
5.6. Summary 31

6. Testing and Calibration **32**
6.1. Excitation Waveform 33
6.1.1. Amplitude 33

6.1.2. Timing . . . 34
6.2. Resistor Calibration . . . 34
6.3. Randles Circuit Test . . . 35
6.4. Salt Solution Test . . . 37
6.4.1. Calibration Line at a Fixed Potential . . . 38
6.5. Ferricyanide Test . . . 39
6.5.1. Peak Height . . . 39
6.5.2. Peak Potential . . . 40
6.5.3. Effect of a Calibration Line . . . 40
6.5.4. Sources of Deviation . . . 41
6.6. Summary . . . 41

7. Environmental and Sustainability Analysis 43

8. Conclusion and Future Work 44
8.1. Conclusion . . . 44
8.2. Future Work . . . 44

References 45

A. Project planning schedule 52

B. Derivations for the System Model 53
B.1. Capacitive Current Rejection in DPV . . . 53
B.2. Negative-Feedback Stability . . . 53
B.3. Control-Loop Feedback Factor . . . 54
B.4. TIA Transfer Function and Compensation . . . 55

C. Display and SD Card Specifications 57

D. Power Management Subsystem Specifications **58**

E. Scan Flow and Timing Diagrams **59**

F. RTIA Calibration Constants and Resistors **60**

G. Webserver Dashboard **61**

H. KiCad Schematic **62**

I. KiCad PCB **63**

J. Case Render **64**

K. AD5941 Capacitor Selection **65**

K.1. Supply Decoupling: 100 nF and 10 µF 65

K.2. Internal Nodes: 470 nF and 4.7 µF 66

L. Bill of Materials **68**

L.1. TIA Noise Gain with a Series Cell 69

List of Figures

2.1. The classic 3-operational-amplifier potentiostat topology, showing the control amplifier, RE buffer, and TIA connected to the three electrodes. 3

2.2. Randles equivalent circuit modeling the electrochemical cell interface. 5

2.3. Waveform profile for DPV. Grey blocks represent the current sampling windows. Where frequency is $f = 1/T_{\text{pulse}}$. Adapted from [1]. 6

3.1. Measurement signal chain of the OpenStat. The the Randles Cell has R_s split at the RE into the part compensated by the control loop, $R_s - R_u$, and the uncompensated resistance R_u between the RE and WE; Z_W is omitted because it only affects the low-frequency response. 12

5.1. Software structure. Arrows show the direction of calls or data. 24

5.2. Top-level program flow: start-up sequence (left) and the cooperative main loop (right). 26

5.3. One DPV step with the default timing. Shaded bands show the two averaging windows. 29

5.4. Automatic selection of the TIA range before a scan. 31

6.1. Excitation waveform measured with the oscilloscope. (a) The complete staircase. (b) One cycle with the measured quantities. (c) Height of every step against 9 and 10 LSB of the AD5941 12-bit DAC. 33

6.2. DPV scans of the dummy cell WE C (left) and Cell 3 (right) from 0.8 V to 1.3 V. 36

6.3. Salt solution measured by each potentiostat. The dotted line marks the potential E^* used for the calibration line of Section 6.4.1. 37

6.4. Calibration line through the five salt points at $E^* = 0.10\,\text{V}$, with the points at all other potentials. 38

6.5. Ferricyanide DPV scans on the same drop, OpenStat run against PalmSens4 run. 39

6.6. The three ferricyanide pairs. (a) Peak height of the OpenStat against the PalmSens4. (b) Potential offset of the OpenStat curve against the peak current. 40

A.1. The decidend Gantt chart for the project planning schedule. 52

B.1. Control loop block diagram for the analog front-end. 54

B.2. Analog front-end transimpedance amplifier circuit. 55

E.1. Flow of one DPV scan. 59

E.2. Step routine. $t_{\mathrm{b}} = T - t_{\mathrm{p}}$ is the base dwell. 59

G.1. The webserver dashboard for the potentiostat. 61

H.1. The KiCad schematic for the potentiostat. 62

I.1. The KiCad PCB for the potentiostat. 63

J.1. The case render for the potentiostat. 64

List of Tables

2.1. Comparison of discrete op-amp and integrated AFE IC implementation approaches. 9

2.2. Comparison of surveyed potentiostat designs. 10

3.1. Phase margin of the TIA loop on the 160 kΩ range ($R_\Sigma = R_L = 110\,\Omega$, $R_u = 0$, GBW = 1 MHz). 14

4.1. Target DPV operating conditions and the requirements they impose. 16

4.2. System requirements. 17

4.3. Capacitors around the AD5941, grouped by the reasoning behind each value. 18

4.4. Required and chosen trace widths for a 10 °C temperature rise in 1 oz/ft^2 external copper. 21

5.1. DPV scan parameters, their default values and permitted ranges. 26

6.1. Measurement conditions. 32

6.2. Measured excitation waveform against the programmed values. 34

6.3. Two-point calibration of the high-speed TIA ranges, recomputed from the recorded readings. Errors are at the +0.25 V check point. 35

6.4. Randles circuit test, 0.8 V to 1.3 V 36

6.5. Salt test, 0.40 V to −0.20 V, all scans on the OpenStat used the 80 kΩ range. r is the correlation coefficient between the PS4 and OpenStat voltammograms of each pair. 38

6.6. Summary of the tests. 41

7.1. Further life-cycle measures. 43

C.1. Key specifications of the DFRobot DFR0928 display [2]. 57

D.1. Main specifications of the power subsystem components. 58

F.1. Current ranges of the high-speed TIA, the calibration constants held in the firmware and the resistor each range was calibrated against. 60

L.1. Bill of materials for the OpenStat prototype. 68

Nomenclature

Acronyms and Abbreviations

ADC	Analog-to-Digital Converter
AFE	Analog Front-End
BLE	Bluetooth Low Energy
BOM	Bill of Materials
CA	Chronoamperometry
CE	Counter Electrode
CV	Cyclic Voltammetry
DAC	Digital-to-Analog Converter
DMM	Digital Multimeter
DPV	Differential Pulse Voltammetry
EIS	Electrochemical Impedance Spectroscopy
GUI	Graphical User Interface
IoT	Internet of Things
LCMS	Liquid Chromatography-Mass Spectrometry
LDO	Low Dropout Regulator
LOD	Limit of Detection
LSB	Least Significant Bit
LSV	Linear Sweep Voltammetry
MCU	Microcontroller Unit
MFT	My Funky Thesis
PS4	PalmSens4
PSRR	Power Supply Rejection Ratio
RE	Reference Electrode
SNR	Signal-to-Noise Ratio
SoC	System-on-Chip
SPE	Screen-Printed Electrode
SWV	Square Wave Voltammetry
TIA	Transimpedance Amplifier
UWED	University of Washington Electrochemical Device
WE	Working Electrode

Definitions

B

Base Dwell The time the potential sits at the base level before a pulse is applied, in each step.

C

Code It is the whole number the firmware writes to the DAC to set its output voltage. The base dwell is the time the potential sits at the base level before a pulse is applied, in each step.

Compliance Voltage The maximum voltage a potentiostat can apply between the counter electrode and the working electrode in order to keep the working electrode at the desired potential

E

Electrochemically Reversible System A system where the electron transfer kinetics between the electrode and the analyte are so fast that they are practically instantaneous compared to the speed of the measurement taken.

V

Voltamogram A plot of current versus applied potential for a given electrochemical reaction

Chapter 1

Introduction

1.1 Background and Motivation

The global landscape of analytical chemistry is currently experiencing a significant shift towards point-of-care medical diagnostics and on-site environmental testing. In South Africa, a critical public health and environmental challenge is the accumulation of active pharmaceutical ingredients, specifically antiretroviral and other antiviral medication, in local water systems [3, 4]. Monitoring these specific pharmaceutical compounds at trace levels requires highly sensitive analytical techniques. Differential Pulse Voltammetry (DPV) was selected as a technique to detect these pharmaceutical compounds in the water due to its high sensitivity [5]. DPV superimposes small pulses of potential onto a linear potential ramp or staircase. By measuring the current difference before and near the end of each pulse, the technique reduces background charging current and achieves high sensitivity [6]. This makes DPV a foundational technique for isolating the minute Faradaic currents required to detect trace pharmaceuticals outside of a laboratory environment [7].

1.2 Problem Statement

The primary barrier when testing water in South Africa is challenging because of the tools required to perform accurate voltammetric analysis are costly and lack in portability [8, 9]. The current commercially available potentiostat, such as the PalmSens4, systems serve as the gold standard for electrochemical analysis. However, these instruments are cost-prohibitive, and operate as closed-source devices [10]. Consequently, it is difficult to integrate them deeply into custom, low-cost or Internet of Things (IoT) architectures [8]. While various open-source alternatives have been developed over the years, many remain tethered to external computers [8] for power and data processing, or rely on proprietary applications that can become obsolete. This presents a distinct need for a smart, autonomous, wireless, and low-cost potentiostat alternative capable of conducting highly sensitive DPV measurements independently in the field [7].

1.3 Project Aim, Objectives and Scope

The primary aim of this project is to design and develop an open-source, low-cost, wireless potentiostat tailored specifically for the DPV detection of antiviral medications in water. The scope of this project encompasses the following hardware and software implementations:

- **Hardware Architecture:** The design uses the AD5941 SoC AFE, which offers a compact physical footprint and ultra-low power consumption for portable electrochemical and impedance-based sensing systems.
- **Microcontroller Integration:** The system is driven by an ESP32-S3 microcontroller, chosen for its dual-core Xtensa LX7 architecture and native Wi-Fi capabilities.
- **Physical User Interface:** The hardware includes a standalone physical user interface consisting of an onboard display, two push buttons, and a rotary encoder.
- **Electrochemical Interface:** The system interfaces with a standard three-electrode setup consisting of a working, reference, and counter electrode. This three electrode setup can be used with a SPE or crocodile clips.
- **Software and Firmware:** Firmware was developed in C++. It enables the system to execute DPV measurements, control the local GUI, and host a web server for seamless data extraction via Wi-Fi.

1.4 Scope Exclusions

The scope of this project focuses on the design, development and analytical validation of the potentiostat hardware and firmware. To validate the DPV measurement performance, laboratory-prepared standard solutions of salt and ferricyanide in a supporting electrolyte were used. These solutions served as benchmarks for the developed potentiostat against commercial instruments such as the PalmSens4. The detection of these compounds in real-world water samples is outside the scope of this project. Including this in the scope would require additional sample preparation, extraction, purification steps to isolate the target analytes and adding a supporting electrolyte [11]. This would introduce additional matrix-interference characterisation and independent analytical confirmation (e.g. via chromatography) that fall beyond the hardware-development engineering focus of this work. The system's performance on real water matrices is identified as a direction for future work.

Chapter 2

Literature

This chapter reviews the literature relevant to the design of a low-cost, portable potentiostat for Differential Pulse Voltammetry (DPV). The purpose of this review is to establish the electrochemical and hardware fundamentals required to design such an instrument, and to survey the existing landscape of commercial and open-source potentiostat designs.

2.1 Electrochemical Sensing

A potentiostat controls the potential between the reference and working electrode while simultaneously measuring the current flowing between the counter and the working electrode. This is a popular three-electrode setup for electrochemical measurement, illustrated in Figure 2.1. The Working Electrode (WE) is the focal area of electrochemical reactions [12]. In a three-electrode system the potential around it is measured and manipulated to study the desired catalyst or analyte, where an analyte is a specific substance or chemical component that is the subject of analysis.

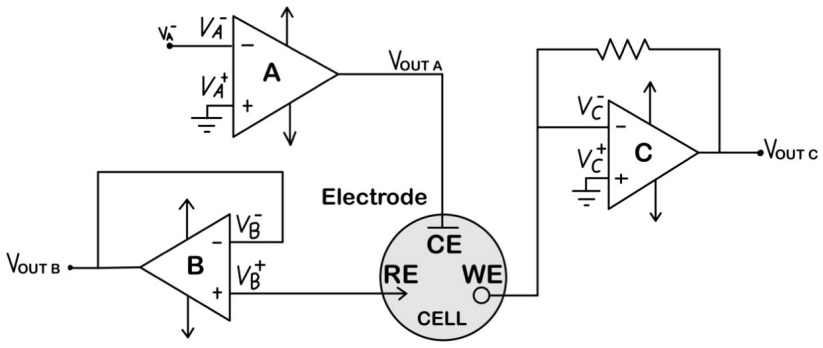


Figure 2.1: The classic 3-operational-amplifier potentiostat topology, showing the control amplifier, RE buffer, and TIA connected to the three electrodes.

The Reference Electrode (RE) provides a stable, known potential that is used as a benchmark to control and monitor the working electrode's potential [13]. Current flow

through the reference electrode is minimal to ensure that its potential remains accurate [13]. The Counter Electrode (CE) (also called the auxiliary electrode) completes the circuit, allowing current to be sourced or sunk between it and the working electrode, which prevents the reference electrode from being affected [13]. The reaction at the counter electrode should be monitored, as corrosion at the electrode could interfere with measurement accuracy [14]. The three-electrode setup is the foundational hardware configuration for the voltammetric techniques discussed in Section 2.1.1, namely Cyclic Voltammetry (CV), Linear Sweep Voltammetry (LSV), Square Wave Voltammetry (SWV), DPV, and stripping voltammetry.

2.1.1 Voltammetry as a Family of Techniques

In any voltammetric measurement, applying a voltage to an electrode generates two types of current:

- **Faradaic current:** The actual current generated by the transfer of electrons during the oxidation or reduction of the target analyte [15]. This is the desired signal.
- **Charging (capacitive) current:** The background current generated simply by charging the electrical double-layer at the electrode surface, which acts like a capacitor caused by the Helmholtz plane [15]. This causes background noise. The double layer is modelled as the capacitance C_{dl} in Section 2.1.2.

The same three-electrode setup handles each of the following techniques in a different way:

- **Linear Sweep (LSV) and Cyclic Voltammetry (CV):** The potentiostat applies a continuous, linear voltage ramp (or a triangular sweep for CV). The counter electrode continuously sinks or sources the resulting Faradaic current.
- **Differential Pulse (DPV) and Square Wave Voltammetry (SWV):** Instead of a continuous ramp, the potentiostat applies a series of discrete voltage pulses or steps. By measuring the current at specific times during the pulse, the system can mathematically subtract the background capacitive (charging) current. This dramatically increases sensitivity, making DPV and SWV the preferred techniques for trace-level detection in biosensing applications.

2.1.2 Equivalent Circuit of the Cell

The two currents described in Section 2.1.1 can be represented by an electrical model of the electrode–electrolyte interface. The widely used Randles equivalent circuit [16], shown in Figure 2.2, models this interface as the solution resistance R_s in series with the double-layer capacitance C_{dl}, which is in parallel with the charge-transfer resistance R_{ct}

and the Warburg impedance Z_W. The charging current flows into C_{dl}, while the Faradaic current flows through R_{ct} and Z_W. R_{ct} represents the resistance to electron transfer at the electrode surface, and Z_W accounts for the diffusion of analyte towards the electrode at low frequencies [17, 18]. The time constant $R_s C_{dl}$ sets how quickly the double layer charges after each potential step, which is central to how DPV rejects the charging current.

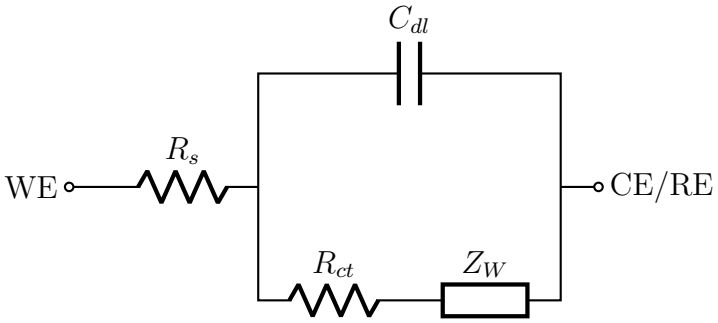


Figure 2.2: Randles equivalent circuit modeling the electrochemical cell interface.

2.1.3 Differential Pulse Voltammetry

DPV was chosen as the focus of this project for its high sensitivity, ability to reject capacitive charging current and strong fit for the intended application [5]. Its excitation waveform is a staircase ramp with a small potential pulse superimposed on each step, as shown in Figure 2.3. The waveform is defined by the step potential, the pulse amplitude and the pulse width [19].

Current is not measured continuously. Two discrete samples are taken per step: I_{before} immediately before the pulse is applied, and I_{end} at the end of the pulse (labelled i_1 and i_2 in Figure 2.3). Their difference,

$$\Delta I = I_{\text{end}} - I_{\text{before}}, \tag{2.1}$$

is plotted against the step potential to form the voltammogram.

The sensitivity of DPV comes from when these samples are taken. After a potential step ΔE, the capacitive current decays exponentially [20],

$$I_c(t) = \frac{\Delta E}{R_s}\, e^{-t/R_s C_{dl}}, \tag{2.2}$$

while the Faradaic current follows the Cottrell equation [17],

$$I_f(t) = \frac{nFA\sqrt{D}\,C^*}{\sqrt{\pi t}}, \tag{2.3}$$

where n is the number of electrons transferred, F is Faraday's constant, A is the electrode area, D is the diffusion coefficient and C^* is the bulk analyte concentration. Because the exponential decays far faster than $1/\sqrt{t}$ [17, Sec. 1.2.4, Sec. 5.2.1], sampling late in the step and late in the pulse allows the capacitive current to decay almost completely, so that $I_{c,\text{before}} \approx I_{c,\text{end}} \approx 0$ and

$$\Delta I \approx I_{f,\text{end}} - I_{f,\text{before}}. \tag{2.4}$$

The full expansion is given in Appendix B.1. In practice this sets a minimum pulse width, since sampling at $t = 5R_sC_{dl}$ leaves less than 1 % of the initial capacitive current ($e^{-5} \approx 0.007$). Because ΔI approximates the derivative of the Faradaic current–potential curve, the voltammogram forms a peak which, for a reversible system, lies half a pulse amplitude away from the half-wave potential $E_{1/2}$ [17].

In continuous sweeps such as CV and LSV, by contrast, charging current is present throughout the measurement. The DPV signal is therefore dominated by Faradaic current, which gives a much higher Signal-to-Noise Ratio (SNR) and lower limits of detection. The cost is reduced time resolution [21].

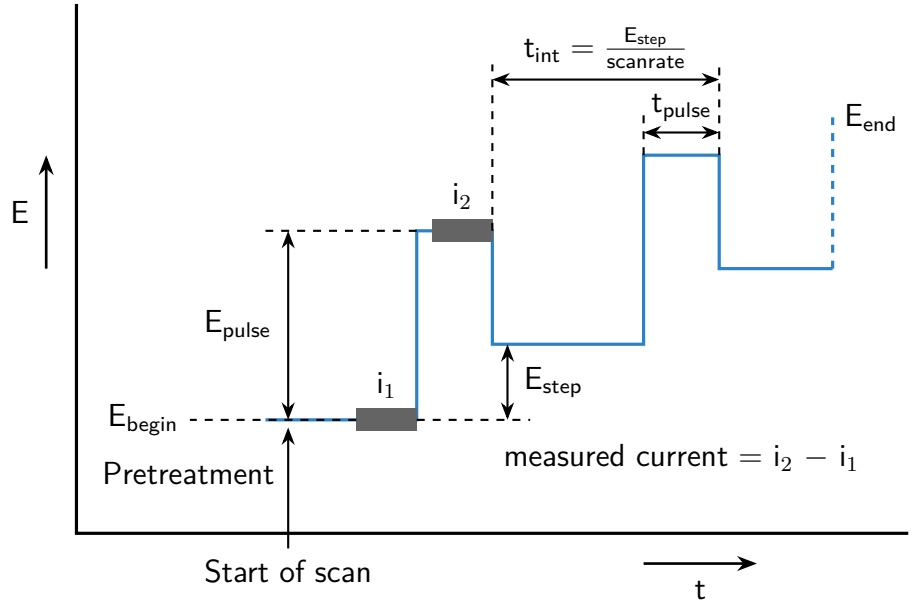


Figure 2.3: Waveform profile for DPV. Grey blocks represent the current sampling windows. Where frequency is $f = 1/T_{\text{pulse}}$. Adapted from [1].

2.1.4 Application to Antiviral Drug Residues in Water

South Africa runs the largest antiretroviral treatment programme in the world, and incomplete removal at wastewater treatment plants releases these drugs into surface water. Efavirenz in particular has been measured at concentrations ranging from ng/L to tens of µg/L in surface water near treatment plant discharges [22], yet South Africa

has no monitoring programme or regulatory limit for these compounds [3]. Existing measurements rely on laboratory chromatography–mass spectrometry (LCMS), which is costly and unsuited to field use.

Target analytes in these applications may be electroactive. These can include antiretrovirals like Efavirenz which undergoes irreversible oxidation and has been quantified by DPV on disposable pencil graphite electrodes [23]. An example of antiviral medication is oseltamivir has been quantified by DPV at modified paste microelectrodes [24] and shows a voltammetric response at gold electrodes [25]. These studies used benchtop instruments and, in most cases, modified electrodes or adsorptive pre-concentration to reach low detection limits. A portable DPV instrument is therefore a necessary step towards on-site screening, although the achievable detection limit ultimately depends on the electrode chemistry as well as the instrument.

2.2 Potentiostat Fundamentals

This section explores the fundamentals of a potentiostat, namely its role, design, performance parameters, and implementation approaches.

2.2.1 The Role of a Potentiostat

The role of a potentiostat is to manage the control amplifier and the TIA, and to buffer the reference electrode. For reference electrode buffering, the electrode must remain perfectly stable at a known half-cell potential. To prevent the voltage from drifting, zero current is allowed to flow through the electrode. This is achieved using an electrometer, an operational amplifier with an extremely high input impedance [26].

The control amplifier drives the system by constantly comparing the desired input voltage, set by a DAC, with the actual voltage measured at the reference electrode. The resulting error signal is minimised by forcing current through the counter electrode and into the solution, which actively compensates for any changes in the electrochemical cell's resistance. The TIA holds the working electrode at a 0 V reference while catching every electron produced or consumed by the reaction [17]. These electrons generate a small Faradaic current that is converted into a potential, which can then be measured by an Analog-to-Digital Converter (ADC) [17]. The precision of the measurements made by the TIA depends not only on the AFEs noise performance but when the sample is taken. As discussed in Section 2.1.3, the capacitive and Faradaic currents are distinguished by their differing decay times. Therefore, to sample accurately, precision timing and specific, predictable sampling windows are required. The occurrence of a timing jitter would directly degrade the signal–to-noise ratio regardless of the AFE's noise performance. To ensure

precision timing a direct requirement is placed on the microcontroller's clock stability and the software architecture that drives the AFE.

2.2.2 Three Operational Amplifier Potentiostat Topology

This section translates the three roles above into a physical circuit architecture, shown in Figure 2.1. The three operational amplifiers are the respective parts of a potentiostat: A is the control amplifier, B is the buffer, and C is the TIA.

2.2.3 Key Performance Parameters

This section establishes the criteria used to evaluate the effectiveness of the hardware, namely current measurement range and resolution, compliance voltage, noise floor, bandwidth, and technique support.

- **Current measurement range and resolution:** In drug residue sensing, very small Faradaic signals are generated. The potentiostat must be able to accurately measure down to the nanoampere (nA) or picoampere (pA) range without clipping or saturating the signal.
- **Compliance voltage:** The control amplifier must drive the cell resistance $R_s + R_{ct}$ of Section 2.1.2. If this resistance is high, the control amplifier needs a high compliance voltage to push the necessary current through the counter electrode [27].
- **Noise floor and Limit of Detection (LOD):** The LOD is fundamentally constrained by the inherent noise floor of the potentiostat's analog front end. In high-sensitivity applications, the TIA requires large feedback resistors to convert picoampere-level Faradaic currents into measurable voltages. However, these resistive components inherently generate thermal (Johnson–Nyquist) noise. Therefore, a device targeting nanoampere or picoampere signals requires a potentiostat topology with an exceptionally low electrical noise floor [28, 29].
- **Bandwidth / scan rate limits:** This is the maximum speed at which the potentiostat can alter the applied potential. Fast techniques such as SWV require high-bandwidth operational amplifiers to avoid signal distortion [17].
- **Technique support:** A system's ability to run CV, DPV, or Electrochemical Impedance Spectroscopy (EIS) depends on its Digital-to-Analog Converter (DAC) resolution and internal waveform generators [30].

2.2.4 Analog Front-End Approaches

Transitioning from the theoretical 3-op-amp topology to a physical implementation requires selecting an appropriate hardware architecture. To combat bulky bench top potentiostats, engineers developed two approaches to AFE's. First, a discrete circuit using individual operational amplifiers. Second, utilising a fully integrated System-on-Chip (SoC) AFE, such as the AD5941. Each approach presents distinct engineering trade-offs regarding power consumption, physical footprint, and architectural flexibility, which are summarised in Table 2.1.

Table 2.1: Comparison of discrete op-amp and integrated AFE IC implementation approaches.

Implementation Approach	**Advantages**	**Trade-offs**
Discrete op-amp design	Architectural flexibility, capable of massive compliance voltages, component-level tuning for extreme precision.	Large physical PCB footprint, high power consumption, higher cost of materials.
Integrated AFE ICs	Millimeter-scale physical footprint, very low power consumption, built-in hardware digital filtering.	Hard-capped compliance voltage ($\leq$ 3 V), fixed internal current measurement ranges, limited deep customisation, requires complex SPI firmware development.

Each approach to AFEs have their strengths and weaknesses, as discussed above. The two approaches are explored in Section 2.3 when surveying existing potentiostat designs.

2.3 Survey of Existing Potentiostat Designs

The democratisation of electrochemical instrumentation has led to a rapid expansion of custom potentiostat designs. This section surveys the architectural evolution of these devices, establishing a performance baseline and identifying gaps in standalone usability, summarised in Table 2.2

Table 2.2: Comparison of surveyed potentiostat designs.

Design / Platform	MCU / AFE Architecture	Primary Techniques Supported	Connectivity & Interface	Onboard Display?	Onboard Analysis?	Open-Source Completeness
Commercial (e.g. EmStat[31])	Proprietary	CV, DPV, SWV, EIS, CA	USB / Proprietary PC GUI	No	No (PC-based)	Closed source
CheapStat [9]	Atmel AVR / Discrete	CV, SWV, CA	None / Basic LCD output	Yes (Basic LCD)	No	Full (PCB/Firmware)
DStat [10]	Atmel XMEGA / Discrete	CV, DPV, SWV, CA	USB / Python PC GUI	No	No	Full (PCB/Firmware/GUI)
Smartphone-Integrated (e.g. UWED [8])	BLE MCU / Discrete	CV, DPV, SWV	Bluetooth / Custom App	No	No	Partial (Hardware focus)
HunStat2 [32]	Mixed / AD5941 SoC	CV, DPV, SWV, EIS	USB / Serial Datalogging	No	No	Full (Hardware/Firmware)
NanoStat [33]	ESP32 / Discrete	CV, SWV, CA	Wi-Fi / Embedded Web GUI	No (Headless)	No	Full (Hardware/Web GUI)

Each design from Table 2.2 is discussed in more detail below, highlighting the evolution of potentiostat architectures and their trade-offs in terms of performance, portability, and usability.

2.3.1 Commercial Instruments

Commercial systems such as the PalmSens4 [34], EmStat [31] and Metrohm Autolab [35] are the benchmark for electrochemical analysis, offering wide dynamic ranges (pA to mA), factory calibration and mature software. However, they are expensive, bulky and closed-source, which prevents their integration into low-cost portable systems. Their value lies in providing the performance reference against which open-source and custom hardware is validated [36]. Open-source designs have therefore largely solved the cost and accessibility problems of commercial instruments, but not yet the combination of high sensitivity and fully standalone operation that field-testing requires

2.3.2 Discrete Open-Source Designs

CheapStat [9] showed that basic analytical electrochemistry was achievable with a Bill of Materials (BOM) under $100, although its 8-bit microcontroller limited ADC/DAC resolution and compliance range. DStat [10] closed much of the performance gap, reaching near-commercial sensitivity with a discrete analog front-end and supporting CV, SWV, DPV and CA through a Python Graphical User Interface (GUI). However, it depends on a USB-connected PC for power and control, which rules out field use. Smartphone-integrated designs such as the UWED [8] achieved portability by moving processing and display to a phone over Bluetooth Low Energy (BLE), but this ties the hardware to a companion app that must be maintained across operating system updates.

2.3.3 AFE-Based and Web-Hosted Designs

Integrated mixed-signal Analog Front-Ends (AFEs) have largely replaced discrete op-amp circuits in recent designs [37]. Systems such as ABE-Stat, HunStat and HunStat2 characterise this generation: using SoCs such as the AD5941, they support techniques up to EIS with sub-milliwatt power consumption, low noise and a small footprint [32, 38]. However, configuring the AFE requires complex register-level firmware, and these devices typically operate as data-loggers with no user interface, relying on raw serial output for analysis [32]. To remove the dependence on a companion app, ESP32-based designs such as NanoStat [33], DM-Stat [39] and SaxStat [40] host an embedded web server. Any browser on any platform can then control the instrument without installed software, at the cost of Microcontroller Unit (MCU) memory [33].

These web-hosted designs are nevertheless headless and cannot be operated without a second device. Early designs such as CheapStat included a basic display, but none of the surveyed systems combines an integrated AFE, a browser interface and a standalone onboard display. Such a dual-interface instrument would allow quick standalone measurements in the field while retaining web-based data access, and this is the gap addressed by this project.

Chapter 3

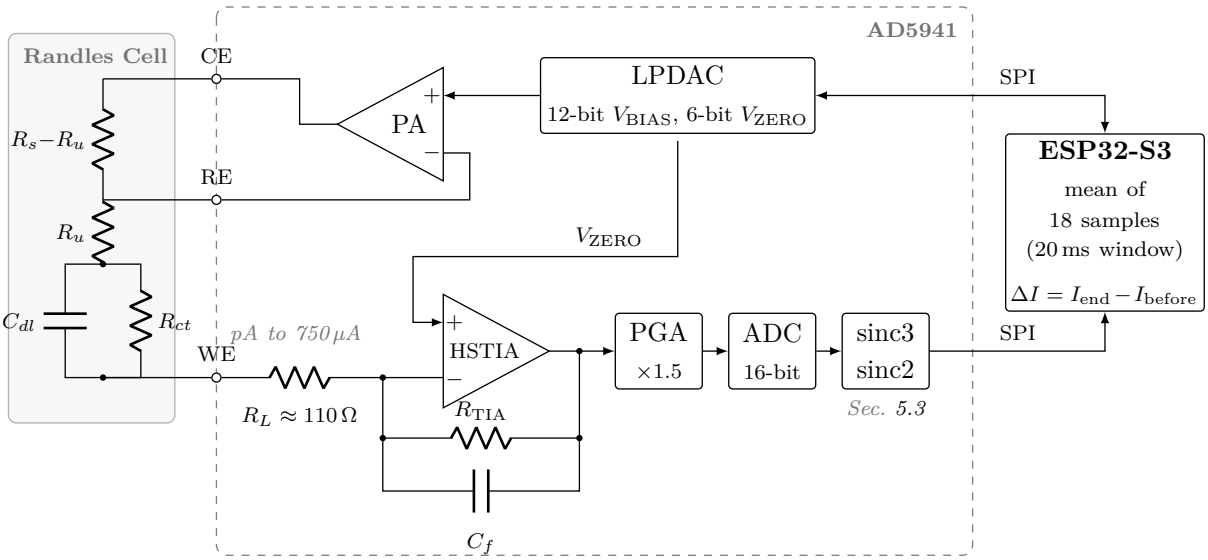
System Modeling & Design

This chapter models the measurement path of the OpenStat, the device this project aims to develop, and derives the requirements it places on the hardware. These requirements describe the choice between the high-speed TIA (HSTIA) and the low-power TIA (LPTIA) of the AD5941, and the conditions under which the chosen amplifier remains stable with an electrode connected.

3.1 Signal Chain

Figure 3.1 shows the measurement path of the OpenStat. The low-power DAC (LPDAC) of the AD5941 sets V_{BIAS} and the potentiostat amplifier which drives the CE until the RE reaches the same potential. The HSTIA holds the WE at V_{ZERO} and converts the cell current to a voltage, which is amplified, digitised and filtered before the ESP32-S3 averages it into I_{before} and I_{end} (Section 5.4).

Figure 3.1: Measurement signal chain of the OpenStat. The the Randles Cell has R_s split at the RE into the part compensated by the control loop, $R_s - R_u$, and the uncompensated resistance R_u between the RE and WE; Z_W is omitted because it only affects the low-frequency response.



In the feeback loop there are two aspects that need to be taken into consideration, the fixed internal load resistor $R_L \approx 110\,\Omega$ [41, pp. 10, 50], and the capacitor C_f across the gain resistor R_{TIA} is programmable from 1 pF to 63 pF [41, p. 52].

3.2 Choice and Operating Limits of the TIA

The AD5941 has two TIAs that can measure the WE current. The LPTIA is intended for continuous, low-power sensing such as gas or glucose sensors, and offers gain resistors up to 512 kΩ [41, p. 10], three times the largest of the HSTIA. The HSTIA is specified for signals up to 200 kHz [41, p. 50], and the datasheet notes that it is the path for high-bandwidth measurements such as ramps and cyclic voltammetry [41, p. 30].

3.2.1 High-Speed or Low-Power TIA

According to the datasheet the recommended configuration for voltammetry is to use the High Speed TIA. This drives the CE from the low-power DAC through the potentiostat loop, disconnects the low-power DAC output capacitors so that the potential can step quickly, and measures the WE current on the HSTIA [41, Table 21, p. 38]. The OpenStat follows this configuration.

The LPTIA was not ruled out on speed advantages would be that with a small feedback capacitor it can also follow a 50 ms pulse, and it has the larger gain resistors. The HSTIA has more practical advantages. Its feedback capacitor C_f is set in firmware [41, p. 52], whereas the LPTIA uses an external capacitor fixed at assembly, so the compensation can be changed without modifying the board. Its bandwidth also covers impedance measurement up to 200 kHz, against 300 Hz for the low-bandwidth loop [41, p. 9], which leaves impedance spectroscopy open as future work.

3.2.2 Stability with a Cell Connected

The AD5941 applies a voltage to the connected cell through feedback. The potentiostat amplifier drives the CE until the RE equals V_{BIAS}, the HSTIA holds the WE at V_{ZERO}, and the current is whatever the cell draws at that potential, up to 750 µA [41, p. 9]. The cell's electrical properties affect how both amplifiers behave. In the control loop, R_u and C_{dl} add a pole at $1/(2\pi R_u C_{dl})$, its gain-bandwidth product is fixed in the AD5941, so this loop cannot be compensated and instead limits the cells that can be driven [17].

The TIA loop can be shaped through C_f. The working electrode is seen from the inverting input of the HSTIA, the cell is C_{dl} in series with $R_\Sigma = R_u + R_L$ to the solution, which the

control loop holds at a fixed potential. Without C_f the noise gain is

$$NG(s) = 1 + \frac{R_{\mathrm{TIA}}}{R_\Sigma + 1/sC_{dl}} = \frac{1 + s\,(R_{\mathrm{TIA}} + R_\Sigma)\,C_{dl}}{1 + sR_\Sigma C_{dl}}, \tag{3.1}$$

with a zero at $f_z = 1/2\pi(R_{\mathrm{TIA}} + R_\Sigma)C_{dl}$ and a pole at $f_p = 1/2\pi R_\Sigma C_{dl}$, above which it levels off at $1 + R_{\mathrm{TIA}}/R_\Sigma$. Unlike the photodiode amplifier often used as a model for this circuit [28], whose noise gain keeps rising, R_Σ isolates C_{dl} at high frequency. The loop crosses over where the open-loop gain of the amplifier, GBW$/f$, has fallen to the noise gain. With no cell connected the noise gain is 1 and the crossover is at the full GBW; with the electrode connected the crossover moves down to the plateau, $f_c \approx \mathrm{GBW} \cdot R_\Sigma/(R_{\mathrm{TIA}} + R_\Sigma)$. For a single-pole amplifier with $f_c \gg f_z$, the phase margin is approximately [42]

$$\phi_{\mathrm{PM}} \approx \arctan\left(\frac{f_c}{f_p}\right), \tag{3.2}$$

so a margin of at least 45° requires $f_c \geq f_p$, or

$$R_\Sigma \geq \sqrt{\frac{R_{\mathrm{TIA}}}{2\pi\,\mathrm{GBW}\,C_{dl}}}. \tag{3.3}$$

The margin is smallest on the largest gain resistor with the smallest series resistance. The datasheet does not give the GBW of the HSTIA, so a conservative 1 MHz is assumed; a larger value only increases the margins. On the 160 kΩ range the plateau is about 1450, so the crossover lies near 1 MHz/1450 ≈ 700 Hz; on the 1 kΩ range it lies near 100 kHz. Table 3.1 gives the margin on the 160 kΩ range with $R_u = 0$, computed from the full loop gain. See Appendix (Appendix X) for the full derivation.

Table 3.1: Phase margin of the TIA loop on the 160 kΩ range ($R_\Sigma = R_L = 110\,\Omega$, $R_u = 0$, GBW = 1 MHz).

C_{dl}	10 nF	100 nF	1 µF	2.5 µF	5 µF
$C_f = 0$	5°	13°	38°	55°	69°
$C_f = 31$ pF (used)	22°	18°	40°	57°	70°
$C_f = 63$ pF	40°	24°	42°	58°	71°

For the C220AT electrode, the typical 10 µF cm^{-2} to 40 µF cm^{-2} of gold [17] gives $C_{dl} \approx$ 1.3 µF to 5 µF [**Check:** replace with measured R_u and C_{dl}]. With the $C_f = 31$ pF used on every range, its margin is at least 44° with no solution resistance and 70° with $R_u = 100\,\Omega$, and the dummy cell WE A (560 Ω + 33 nF) of Section 6.3 has 53°. The weak case is about 100 nF, such as a microelectrode, where the margin falls below 20°; raising C_f to 63 pF on the sensitive ranges would help.

3.2.3 Analog and Digital Bandwidth

The measurement path of the AD5941 has two bandwidths with opposite purposes. The analog path must be wide so that the cell current is converted faithfully as it changes at each pulse edge. The HSTIA is specified for signals up to 200 kHz [41, p. 50], corresponding to a time constant below 1 µs. The digital path is then made narrow to reject noise. The ADC samples at 800 kS/s [41, p. 55], and two averaging (sinc) filters reduce this in stages: the sinc3 filter averages five samples, lowering the rate to 160 kS/s with a bandwidth of about 42 kHz, and the sinc2 filter averages a further 178, lowering it to about 900 S/s with a bandwidth of about 290 Hz. Because the narrowing happens after the signal has been digitised, it lowers the noise of each sample without slowing the response to a pulse edge.

3.3 Summary

The switchable ranges for are needed to span tens of picoamperes to the maximum 750 µA, the HSTIA was used in the datasheets's voltammetry configuration, as the LPTIA's larger resistors offer no benefit. With R_L in series with the cell, the TIA is stable with the test electrode on every range.

Chapter 4

Hardware Design

This chapter details the hardware design focused on ensuring that the prototype functions correctly and to an adequate standard in its physical realisation by justifying the hardware selected. A primary concern was minimising the noise coupled into the measurement from other parts of the system, such as the Wi-Fi transmissions of the ESP32-S3, the display, and reads and writes to the SD card. This was addressed through decoupling, a ferrite bead on the analog supply, and the separation of the analog and digital grounds.

4.1 AFE Requirements

The research group at the University of Johannesburg [43] specified the measurement method as well as the parameters that the OpenStat should fulfill. The operating requirements are listed in Table 4.1. The same parameters are used on the PalmSens4 reference instrument in Chapter 6.

Table 4.1: Target DPV operating conditions and the requirements they impose.

Parameter	**Value**	**Requirement on the device**
Potential window	0.8 V to 1.3 V	Potential covers 0.8 V to 1.325 V, including the pulse
Step potential	5 mV	DAC LSB $\leq$ 1 mV; each level within $\pm$0.5 LSB
Pulse amplitude	25 mV	25$\pm$1 mV
Pulse time	50 ms	Timing error changes the sampled current by $\leq$ 2 %
Interval time	0.5 s	0.5 s $\pm$ 1 % per step, held over the full scan
Current range	$\pm$750 µA	No saturation at $\pm$750 µA; gain error $\leq$ 1 % per RTIA range after calibration

4.2 System Requirements

The table below details the requirements that the OpenStat must meet.

Table 4.2: System requirements.

ID	Requirement	Rationale
Processing and memory		
R1	Buffer a complete scan in memory	Samples are written to storage only after the scan finishes
R2	At least two SPI buses available for peripherals	One dedicated to the AD5941, one shared by the display and SD card
R3	Enough GPIO for all peripherals	AD5941, display, SD card and user inputs
Display and storage		
R4	Colour display with an SPI or I^2C interface	Compatible with the microcontroller
R5	Display fits in landscape orientation	Handheld form factor
R6	Sufficient display resolution	Current–voltage plot displayed
R7	Non-volatile storage	Field measurements accessed later
User interface		
R8	Standalone operation without a computer	Scan parameters set and scans started in the field
Power		
R9	Battery powered	Portable field use
R10	Rechargeable from USB	No battery replacement needed
R11	Regulated supply over battery discharge range	Cell voltage varies around 3.0 and 4.2 V
R12	Battery life of a day	24 hour battery life

To start implementing the above requirements, the microcontroller is specified next.

4.3 Microcontroller

The range of microcontroller options is vast, the ESP family was selected as it holds a strong reputation for native Wi-Fi, Bluetooth, sufficient processing power and cost-effectiveness, making it an easy choice. The ESP32-S3-WROOM-2-N32R16V was selected for its 32 MB of flash and 16 MB of PSRAM [44]. The larger flash and PSRAM give more flexibility when handling large assets and when running several functions of the device at once. Its dual-core processor allows large tasks to be managed while keeping operation smooth, and its Wi-Fi capability is used to host the web server described further in Section 5.1.

4.4 PCB Component Placement and Layout

The key to low noise in the AD5941 is sufficient filtering and decoupling, together with correct component placement. The rules followed when selecting the decoupling capacitors were taken from the Analog Devices MT-101 tutorial [45]. Finally, a ferrite bead is placed in series with the supply pin. This keeps external high-frequency noise away from the IC and prevents internally generated noise from affecting the rest of the system [46].

4.4.1 Capacitor Choices

The capacitors around the AD5941 can be separated into two groups, decouping the external supply pins and connecting internal nodes with the IC. The Table 4.3 summarises the capacitors choices for their respective roles. The supply-pin values follow the MT-101 two-tier convention [45] and are checked against a target supply impedance, while the internal-node values follow the datasheet [41]. The full derivations for capacitor choices are given in Appendix K.

Table 4.3: Capacitors around the AD5941, grouped by the reasoning behind each value.

Pin	Value	Role and justification
DVDD (6), ×2 IOVDD (26) AVDD (30) AVDD (41)	100 nF	*High-frequency supply decoupling.* No datasheet value given, so one ceramic capacitor per supply pin, as per MT-101 [45]. Stays below the 1 Ω target impedance to beyond 100 MHz (Eq. (K.4)) [47].
AVDD (30)	10 µF	*Bulk decoupling.* Second tier of MT-101 [45]. Meets the 1 Ω target above 15.9 kHz (Eq. (K.2)) and forms an LC filter with the ferrite bead (Section 4.4.2).
`VREF_2V5` (4) `VBIAS_CAP` (31) `DVDD_REG_1V8` (14) `AVDD_REG` (5)	470 nF	*Reference and internal LDO decoupling.* Datasheet value [41, pp. 11, 20–21], set by the stability of on-chip buffers and regulators [48]. No value is given for `AVDD_REG`, so it is matched to `DVDD_REG_1V8`.
`VREF_1V82` (43)	4.7 µF	*ADC reference reservoir.* The datasheet recommends 4.7 µF [41, p. 19] but tests at 0.47 µF [41, p. 11]. The larger value was chosen as the more conservative.
`VBIAS0` (8) `VZERO0` (9)	100 nF	*Low-power DAC output noise filter.* Datasheet value [41, p. 20].
`RESET` (22)	100 nF	*Reset debounce.* With the 10 kΩ pull-up, $\tau = 1\,\text{ms} \gg 1\,\mu\text{s}$ minimum reset pulse [41, p. 14].

Together, the two tiers of supply capacitors hold AVDD below the 1 Ω target from about 16 kHz to beyond 100 MHz, which keeps supply noise below one LSB at the ADC reference.

4.4.2 Ferrite Bead

AVDD shares the 3.3 V rail with the ESP32-S3, display and SD card. These components place noise on the rail from tens of kilohertz to hundreds of megahertz. This noise comes from the SPI clocks, regulator ripple and current bursts. A ferrite bead in series with AVDD forms a low-pass filter with the AVDD decoupling capacitors, and over its resistive band it dissipates this noise as heat instead of reflecting it [46]. A bead was preferred over a separate regulator for AVDD because it adds a single small, inexpensive part that is most effective at high frequencies. The Murata BLM18AG601SN1D was chosen for its high impedance of 600 Ω at 100 MHz, with a rated current of 500 mA and a maximum DC resistance R_{DC} of 0.38 Ω [49].

The bead operates far from the point where DC current degrades its filtering. Impedance curves form the datasheet are measured at zero bias, and under load the core saturates and the inductance falls. Therefore the beads should operate at no more than about 20 % of its rated current [46]. The worst-case AVDD current of about 10 mA is 2 % of the rating, so the datasheet impedance applies. The resulting 3.8 mV drop uses 1.3 % of the 0.3 V allowed between AVDD and DVDD [41, p. 7].

The primary risk with ferrite beads is resonance. Below its crossover frequency the bead is almost purely inductive, and it resonates with the decoupling capacitors. An underdamped resonance amplifies noise instead of attenuating it, which is called peaking [46]. For the series RLC filter,

$$f_0 = \frac{1}{2\pi\sqrt{L_{\mathrm{bead}}C}} \approx 50\,\mathrm{kHz}, \qquad Q = \frac{1}{R_{\mathrm{DC}}}\sqrt{\frac{L_{\mathrm{bead}}}{C}} \approx 0.82, \tag{4.1}$$

with $C = 10\,\mu\mathrm{F}$ on AVDD and an assumed $L_{\mathrm{bead}} \approx 1\,\mu\mathrm{H}$, consistent with the 600 Ω rating at 100 MHz. Peaking occurs only for $Q > 1/\sqrt{2}$ [42].

It is the 10 µF bulk capacitor that keeps the resonance damped. With only the two 100 nF capacitors, Q would be about 5.9, and the filter would peak by 15 dB near 360 kHz. The large capacitor lowers $\sqrt{L/C}$ until the bead's own series resistance is sufficient to damp the filter.

At high frequencies the filter performs as intended. At 100 MHz the bead's 600 Ω and the 0.63 Ω of the local capacitors (Eq. (K.4)) form a divider that attenuates noise by about $20\log_{10}(0.63/600) \approx -60\,\mathrm{dB}$.

4.4.3 Analog and Digital Ground

The PCB carries two types of signals, analog and digital, whose requirements conflict. The AD5941 AFE measures small Faradaic currents, so any disturbance appears as a measurement error. In contrast, the ESP32-S3, the display and the user inputs are digital. This means that they draw fast pulses of current at set logic levels, and that current must return to its source through ground. A ground plane is not ideal for this as it has a small resistance and inductance, so a rapidly changing current induces a voltage according to $v = L\,\mathrm{d}i/\mathrm{d}t$. If the analog and digital circuits share the same ground return, this voltage is added to the analog reference and the digital switching noise is superimposed on the measured signal [50].

To combat this, the PCB was partitioned into an analog region and a digital region, which are the lower and upper parts of the board respectively, with all digital components and traces confined to the digital region. Partitioning is effective because, at high frequencies such as those in digital edges, the return current follows the path of least inductance and flows in the ground copper directly beneath its own signal trace [51, 52]. In accordance with Analog Devices guidance, the analog and digital grounds should be joined through the shortest possible connection [50, 52]. This is done by stitching them together directly beneath the AD5941, commonly called a star ground connection [50]. The stitched bridge also serves as the crossing point for the SPI bus and the reset and interrupt lines, so each signal retains a continuous return path directly underneath it. If the two grounds been fully separated and joined only at a distant point, such as the power input, these return currents would be forced around a large loop that acts as an antenna [53].

4.4.4 Trace Widths

Trace widths were verified using the IPC-2221 curve-fit equations [54] as implemented in the DigiKey trace width calculator [55]. The required cross-sectional area A (in mil^2) for a current I (in A) and an allowed temperature rise T_{rise} (in °C) is given by Equation (4.2), and the corresponding trace width W (in mil) for a copper thickness t (in oz/ft^2) by Equation (4.3):

$$A = \left(\frac{I}{k\, T_{\mathrm{rise}}^{b}} \right)^{1/c} \tag{4.2}$$

$$W = \frac{A}{t \times 1.378} \tag{4.3}$$

The PCB is a two-layer board, so all traces are on external layers, for which $k = 0.048$, $b = 0.44$ and $c = 0.725$ [55]. A copper thickness of 1 oz/ft^2, an ambient temperature of 25 °C and a temperature rise of 10 °C were used. The required and chosen widths are given in Table 4.4.

Table 4.4: Required and chosen trace widths for a 10 °C temperature rise in 1 oz/ft^2 external copper.

Net	Design current	Required width	Chosen width	Capacity of chosen width
AD5941	10 mA	0.0005 mm (0.021 mil)	0.3 mm (11.8 mil)	1.00 A
Display	33 mA	0.0027 mm (0.107 mil)	0.3 mm (11.8 mil)	1.65 A
ESP32	1 A	0.3 mm (11.8 mil)	0.6 mm (23.6 mil)	1.65 A

4.5 Display and SD Card

The DFRobot Fermion 1.8" TFT (DFR0928) was selected as the display because it meets requirements R4–R7 from Table 4.2, including the addition of an onboard SD card slot. Additional information regarding the display is given in Appendix Table C.1.

The display and the SD card share a single SPI bus. The main driving factor in this decision, was the selection of the N32R16V as it only has two SPI lanes [44], which makes combining the SD card and display a requirement. This allows a full SPI bus to be dedicated to the AD5941, so that scans are minimally affected. This does come with trade-offs, that should be acknowledged such as the inability to draw to the screen and to write to the SD card simultaneously. This is avoided by buffering the samples in PSRAM and writing the file only after the scan has finished.

4.6 Power Management

The power subsystem was chosen to meet the requirements of full-day battery life, over-voltage and under-voltage protection, and affordable components. The chosen components are the TP4056 charger with a DW01A protection IC, the Pololu S9V11E2F5 step-up/step-down regulator producing 5 V, and a 5000 mA h, 3.7 V single-cell LiPo battery. Their main specifications are summarised in Table D.1.

4.6.1 Battery

To verify that the selected 5000mAh battery fulfills the full-day operational requirements, the systems current was measured during idle, transmission over WiFi and DPV measurements. The supply current was measured by powering the PCB at 5 V from a Siglent SPD3303C bench supply and reading the output current from the supply. The SPD3303C has a current readback resolution of 10 mA and an accuracy of ±(0.5 % of reading+2 digits) [56], that is, about ±21 mA at these currents. The board draws 100 mA ± 21 mA at idle and 160 mA ± 21 mA while transmitting over Wi-Fi. The idle current did not change when taking a DPV measurement. Taking 160 mA as the worst case, the power drawn from the

battery through the regulator ($\eta_{\text{reg}} \approx 0.9$) is

$$P_{\text{bat}} = \frac{5\,\text{V} \times 160\,\text{mA}}{\eta_{\text{reg}}} = \frac{0.80\,\text{W}}{0.9} \approx 0.89\,\text{W}, \tag{4.4}$$

and the run time is

$$t_{\text{run}} = \frac{E_{\text{bat}}\,\eta_{\text{reg}}}{V_{\text{out}}\,I_{\text{load}}} = \frac{5\,\text{A h} \times 3.7\,\text{V} \times 0.9}{5\,\text{V} \times 160\,\text{mA}} \approx 21\,\text{h}. \tag{4.5}$$

Allowing for the measurement uncertainty, the run time with Wi-Fi transmitting lies between 18 h and 24 h, and at idle between 28 h and 42 h. Even the lower bound comfortably meets the R12 as well as R9-R11.

4.6.2 Charging

The TP4056 is a linear constant-current, constant-voltage charger with a 4.2 V float voltage. In its current configuration the charging current is 1A [57]. The last stage of the charging cycle slows down, so a factor k is introduced to account for the longer charge time. From timing the charging process, the average charge time was found to be about 6.5 h, hence $k = 1.3$:

$$t_{\text{charge}} = k \times \frac{C_{\text{bat}}}{I_{\text{charge}}} = 1.3 \times \frac{5\,\text{A h}}{1\,\text{A}} \approx 6.5\,\text{h}. \tag{4.6}$$

4.6.3 Regulation

The Pololu S9V11E2F5 covers the full battery voltage range with sufficient output current, but it requires the addition of an external low-voltage cutoff. Its 700 mA start-up current limit is sufficient for the ESP32-S3 to boot, and its soft-start ramps the output up gradually [58]. The regulator has a 100 kΩ pull-up resistor from its enable pin to V_{IN}; the enable pin turns the regulator off below 0.7 V and on above 0.8 V [58]. Connecting a 33 kΩ resistor from the enable pin to ground forms a divider that switches the regulator off at about 2.8 V and back on at about 3.2 V, as given in Equation (4.7).

The divider draws about 28 µA continuously, which is sufficiently low for this application.

$$V_{\text{off}} = 0.7\,\text{V} \times \frac{100\,\text{k}\Omega + 33\,\text{k}\Omega}{33\,\text{k}\Omega} \approx 2.8\,\text{V}, \qquad V_{\text{on}} = 0.8\,\text{V} \times \frac{133\,\text{k}\Omega}{33\,\text{k}\Omega} \approx 3.2\,\text{V}. \tag{4.7}$$

A design oversight was the choice to boost the voltage to 5 V when no component requires a 5 V supply. A 3.3 V step-up/step-down regulator would have been more efficient. The large 5000 mA h battery makes this loss affordable, although it is not ideal.

4.6.4 Input Decoupling

No single capacitor has a low impedance across the whole frequency range, so a large capacitor for low frequencies is paired with a small one for high frequencies [45]. Decoupling is required because high-frequency noise couples through to the regulator output and must be kept out of the chip [45]. A 100 µF bulk capacitor is used at the output of the boost regulator, which is the input to the PCB, to act as a local charge reservoir. An additional 100 nF capacitor shunts high-frequency noise away from the IC [45].

4.7 User Interface

The interface consists of two push buttons and a rotary encoder with an integrated push button. All are wired in an active-low configuration with 10 kΩ hardware pull-up resistors. A capacitor on each line forms an RC filter with its pull-up, so on release the pin only reads high once the capacitor has charged past the ESP32-S3 input-high threshold of $0.75 \times V_{DD}$ [59]. This takes $t = RC \ln(4) \approx 1.39RC$.

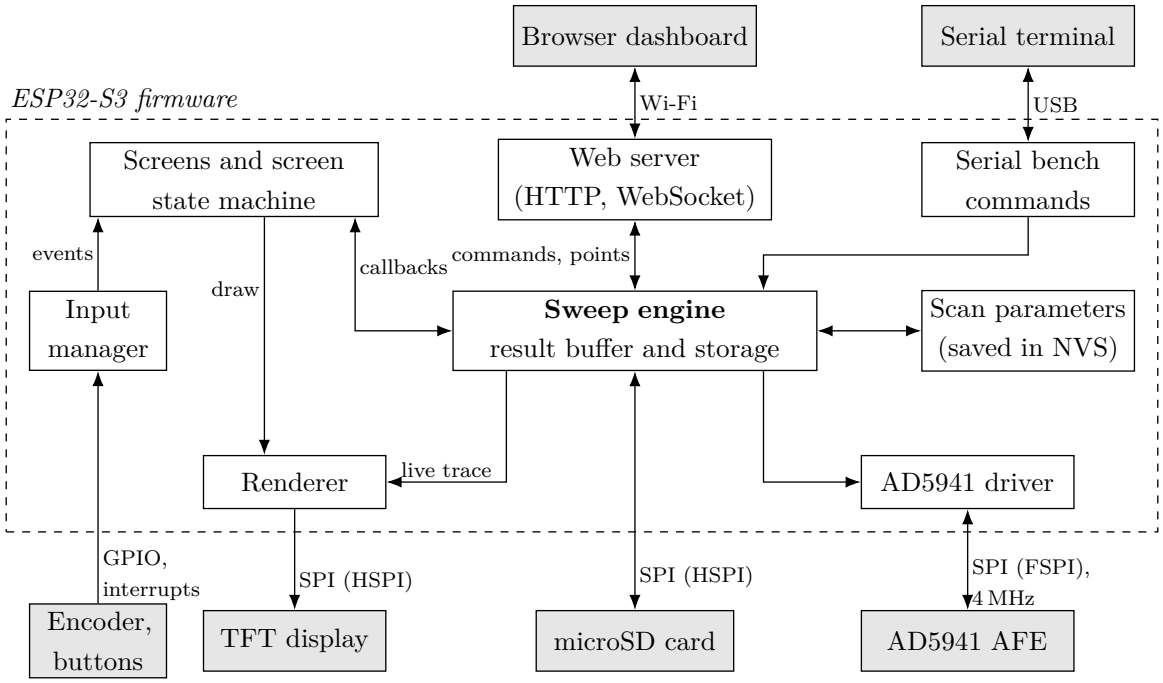
The push buttons use 100 nF ($\approx$ 1.4 ms), which is imperceptible to the user but long enough to absorb most contact bounce [60]. The encoder lines use 10 nF ($\approx$ 0.14 ms), because the filter must not smear the quadrature edges during fast rotation. The EC11E bounces for up to 2 ms [61], which is longer than the edge spacing allows even at moderate speeds. A hardware-only debounce is therefore not strictly possible, but was done for good practice. The RC filter suppresses short spikes, and the quadrature decoder rejects the remaining bounce in software.

Chapter 5

Software Design

The hardware described in Chapter 4 has the ability to apply a potential to the cell and measure the current that flows, but it is the firmware that turns these two features into a useful measurement. This chapter describes the structure of the firmware and the logic of how a measurement is performed.

Figure 5.1: Software structure. Arrows show the direction of calls or data.



Each of the modules in Figure 5.1 is described in the sections that follow.

5.1 Software Overview

The firmware is written in C++ on the Arduino framework for the ESP32-S3 and is built with PlatformIO. Third-party libraries like Adafruit GFX and ST7735 are used for the display, ESPAsyncWebServer and AsyncTCP for the web server and ArduinoJson for JSON encoding.

There were three design decisions shaped the structure of the software:

- **Project-specific AD5941 driver.** Analog Devices supplies a general-purpose software library for the AFE, which covers every measurement mode of the device. The driver for this project was adapted from this library and was written around the needs of this project. This gives the firmware direct control of the gain resistor R_{TIA} of the transimpedance amplifier, which the automatic range selection in Section 5.5.2 switches repeatedly before a scan.
- **One sweep engine, two front ends.** The on-device screens and the web dashboard both start, stop and observe a measurement through the same sweep engine, so a scan behaves identically regardless of the interface that started the sweep.
- **A dedicated SPI bus for the AFE.** The display and the SD card share an SPI bus so display and storage traffic never occupies the bus used for the measurement.

The dedicated bus keeps display and storage traffic away from the measurement in hardware, but all of these modules still share the processor's time. How that time is divided, and when the measurement takes priority, is set by the top-level program flow.

5.2 Top-Level Program Flow

All project code runs from a single loop, shown in Figure 5.2. During boot-up the device does a check to see if the WiFi, SD card and AD5941 is working correctly, after these checks are done the user is taken into the home display. The state of the device is stored in three places, the active screen, a flag that records whether a scan is running, and the phase of the DPV step inside the AFE driver. The loop services each of these in turn on every pass and none of them blocks the loop, except when the device is sampling which is discussed in Section 5.4.

5.3 DPV Measurement Sequence

A scan is defined by the parameters required in Table 5.1. They are stored in non-volatile storage (NVS), a partition of the ESP's flash memory, edited on the device and restricted to the limits seen in Table F.1 whenever they are loaded or edited to protect the hardware. The scan rate is not set directly and is found by dividing the number of steps by the period.

Figure 5.2: Top-level program flow: start-up sequence (left) and the cooperative main loop (right).

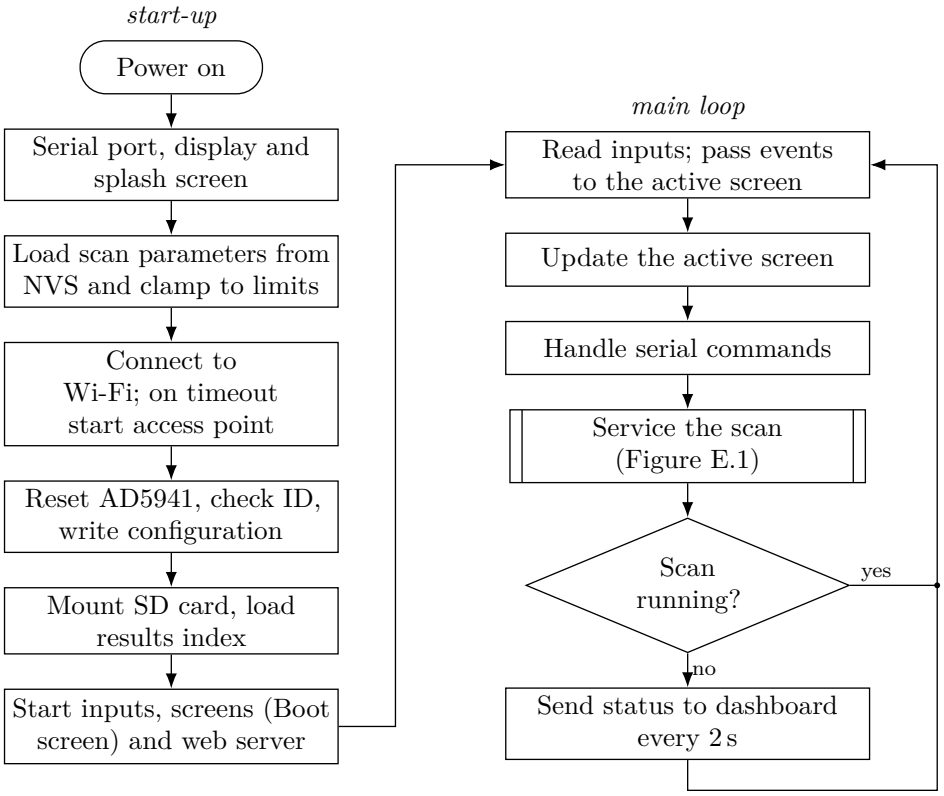


Table 5.1: DPV scan parameters, their default values and permitted ranges.

Parameter	Symbol	Default	Permitted range
Start potential	E_{start}	0.80 V	-0.35 V to $E_{\text{end}} - \Delta E_{\text{s}}$
End potential	E_{end}	1.30 V	$E_{\text{start}} + \Delta E_{\text{s}}$ to 1.30 V
Step	ΔE_{s}	5 mV	1 mV to 50 mV
Pulse amplitude	ΔE_{p}	25 mV	-100 mV to 100 mV
Pulse width	t_{p}	50 ms	10 ms to 500 ms
Pulse period	T	500 ms	50 ms to 5000 ms
TIA range	R_{TIA}	automatic	automatic, or one of eight fixed values

5.3.1 Generating the Potential

The cell potential is set by the two outputs of the low-power DAC of the AD5941. The 6-bit output V_{ZERO} biases the high-speed transimpedance amplifier (TIA), which holds the working electrode at that voltage, and is fixed at 1.7125 V. The 12-bit output V_{BIAS} drives the reference electrode through the potentiostat amplifier and carries the staircase and the pulses.

The applied potential is therefore

$$E = V_{\mathrm{ZERO}} - V_{\mathrm{BIAS}}, where \qquad V_{\mathrm{BIAS}} = 0.2\,\mathrm{V} + D_{12} \times \frac{2.2\,\mathrm{V}}{4095}, \tag{5.1}$$

and D_{12} is the 12-bit code and one code is 537.2 µV which represents the voltage change per code. With V_{ZERO} fixed at this value, the permitted range of E in Table 5.1 keeps V_{BIAS} between 0.41 V and 2.06 V, inside the 0.2 V to 2.4 V output range of the DAC.

5.3.2 Scan Flow

Before any timing starts, the TIA range is selected (Section 5.5.2) and the requested timing is checked against the minimum that the sampling scheme can support (Section 5.4), a scan that fails this check is stopped. The scan consists of

$$N = \left\lfloor \frac{E_{\mathrm{end}} - E_{\mathrm{start}}}{\Delta E_{\mathrm{s}}} \right\rfloor \tag{5.2}$$

points, and the end potential itself is not applied as a base level. Each point is produced by the step routine in Figure E.2, which the main loop calls on every pass. The routine returns immediately unless a sampling window is due, so the user interface stays responsive during the 450 ms base dwell.

Points are held in memory during the scan and written to the SD card as a single file only once the scan has stopped, so that card activity cannot disturb the measurement. Every way of leaving a scan, whether completion, a stop request or an overload, passes through the same stop routine, which stops the ADC, returns the cell to $E = 0\,\mathrm{V}$ and saves the points captured so far. For more detail Figure E.1 in Appendix **??** shows the process of how one scan is run.

5.4 Pulse Sampling and Timing

DPV rejects the capacitive charging current only if I_{before} and I_{end} are each sampled after the transient from the preceding edge has decayed, as shown in Section 2.1.3. When the current is sampled, and for how long, therefore sets both the noise of the measurement and how much charging current leaks into ΔI.

5.4.1 Software Timing

All DAC updates and ADC conversions are done on the microcontroller. The staircase and the pulses are individual DAC writes over SPI, scheduled by the ESP32-S3 against its millisecond clock, and data-ready is detected by polling the interrupt flag register of the AD5941. This was a deliberate choice: timing the measurement in firmware gives

the ESP32-S3 full control over when every operation in the system takes place, so the measurement can be scheduled around the display, storage and network activity instead of running alongside it.

That control is used to give the measurement priority over everything else while a scan is running. Whenever the current is being sampled, the firmware does nothing but sample: the buttons are not read, the display is not updated and nothing is written to the SD card. This keeps the digital activity of the device out of the sampling windows, which reduces the noise coupled into the measurement and improves its accuracy. The cost is that the timing between windows depends on the main loop, and the sampling scheme below was designed so that this dependence does not reach the measured current.

5.4.2 Averaging Window

Before the ADC of the AD5941 transmits the measruements to the ESP32-S3 its output passes through two digital filters inside the device, a sinc3 filter followed by a sinc2 filter [41, p. 57]. A sinc filter is a low-pass filter that averages a block of consecutive samples and outputs one result per block, so it both suppresses noise and lowers the data rate. The number of samples per output is the oversampling ratio. In the configuration used, the sinc3 filter reduces the raw 800 kHz ADC rate by a factor of 5 to 160 kHz, and the sinc2 filter reduces it further by a factor of 178 [41, p. 59], so the AD5941 produces one filtered result every $T_\mathrm{s} = 178/160\,\mathrm{kHz} = 1.1125\,\mathrm{ms}$. The device also offers a 50 Hz/60 Hz notch filter after the sinc2 filter, but with it enabled the settling time after a step is 37 ms, against 2.245 ms without it [41, p. 17]. A 37 ms settling time would consume most of a 50 ms pulse, so the notch filter is bypassed and mains rejection is provided in firmware instead: each current value is the mean of $N_\mathrm{w} = 18$ consecutive sinc2 results. The window length is

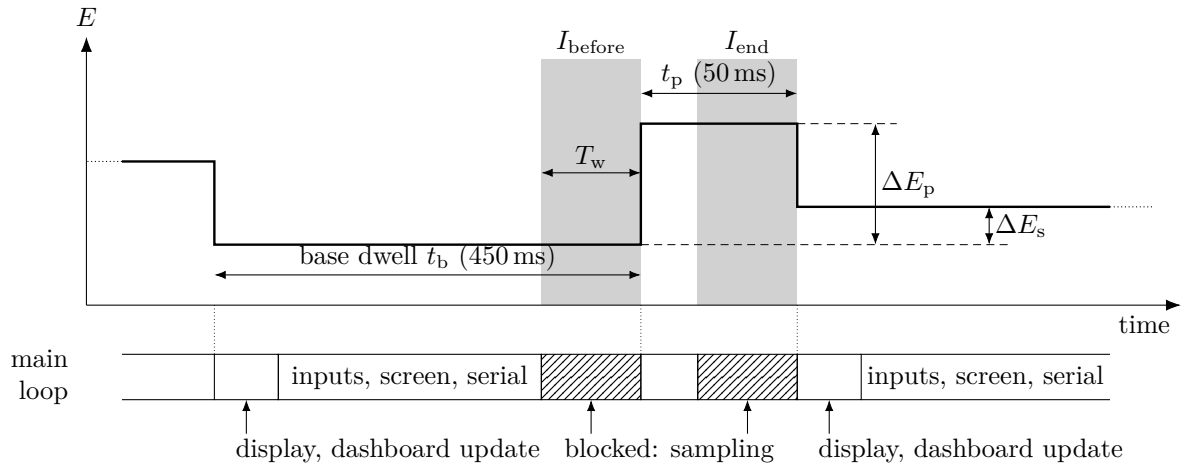
$$T_\mathrm{w} = N_\mathrm{w} T_\mathrm{s} = 18 \times 1.1125\,\mathrm{ms} = 20.025\,\mathrm{ms}, \tag{5.3}$$

which is one period of the 50 Hz mains to within 0.13 %. This places zeros at mutiples of 49.95Hz creating an effective filter. The window is defined by a sample count and not by a duration, so every window in a scan has exactly the same length, and the ADC runs continuously for the whole scan so that the filters are never restarted between windows.

5.4.3 Placing Windows

Figure 5.3 shows where the two windows fall within a step. Each window starts 21 ms before the edge that follows it, so that it ends at the edge: I_before is the mean current over the last 20 ms of the base dwell and I_end the mean over the last 20 ms of the pulse. These are the points of each phase at which the charging current has had the longest time to decay.

Figure 5.3: One DPV step with the default timing. Shaded bands show the two averaging windows.



Three properties of this scheme limit the effect of software timing:

- **During sampling the loop is blocked.** While a window is being sampled, the firmware waits on the data-ready flag of the ADC and does nothing else. This means that the 18 samples are consecutive, and no activity takes place while the current is being sampled.
- **Lateness only adds settling time.** If a window starts late, the sampling starts later after the edge, so loop delay lengthens the pulse slightly but cannot move a window into the charging transient.
- **Slow work is placed after the step.** The display and the dashboard are updated when a point is returned, which is at the start of the next base dwell.

A scan is refused if the pulse width or the base dwell is shorter than the window plus a 5 ms margin, that is, 25 ms. At this minimum a window still starts 4 ms after the preceding edge, which is longer than the 2.245 ms settling time of the filter.

5.4.4 Limitations

Each phase is timed from the end of the previous window and there is no correction against an absolute schedule, so the real step period is slightly longer than T and the total scan time is slightly longer than NT. The buttons are not read during a window, so a stop request can be delayed by up to 20 ms. Commands from the dashboard are handled in the callback context of the web server library and not in the main loop, so their timing relative to a window is not controlled.

5.5 Current Measurement and Ranging

The ADC measures the differential output of the high-speed TIA through its gain stage, set to a gain of 1.5. The specific gain value was selected as it is the only factory-calibrated gain from the datasheet [41, p. 56] and is appropriate for this configuration. The mean code $\bar{D}$ of a window is converted to the TIA output voltage with the transfer function given in the datasheet for this gain [41, p. 56], and then to a current using the nominal value of the selected gain resistor:

$$V_{\mathrm{TIA}} = \frac{1.835\,\mathrm{V}}{1.5} \times \frac{\bar{D} - 2^{15}}{2^{15}}, \qquad I_{\mathrm{raw}} = \frac{V_{\mathrm{TIA}}}{R_{\mathrm{TIA}}}. \tag{5.4}$$

A gain and an offset correction, held in a table with one entry per range k, are then applied:

$$I = G_k\, I_{\mathrm{raw}} + O_k. \tag{5.5}$$

Both samples of a step are taken on the same range, so the offset cancels in the differential current and only the gain affects the result:

$$\Delta I = I_{\mathrm{end}} - I_{\mathrm{before}} = G_k \left(I_{\mathrm{raw,end}} - I_{\mathrm{raw,before}}\right). \tag{5.6}$$

5.5.1 Ranges and Calibration

The eight gain resistors of the high-speed TIA give the ranges from 200 to 160 kΩ. The full-scale current follows from the ±0.9 V input range of the ADC at this gain [41, p. 8], and the resolution is the current that corresponds to one ADC code before averaging.

The potentiostat amplifier that drives the counter electrode can source or sink only ±750 µA in its normal mode [41, p. 9], which is the mode used. The 1 kΩ resistor covers this range up to its overload threshold of 720 µA (Section 5.5.2). The 200 Ω range would only become useful with the high-current mode of the amplifier enabled.

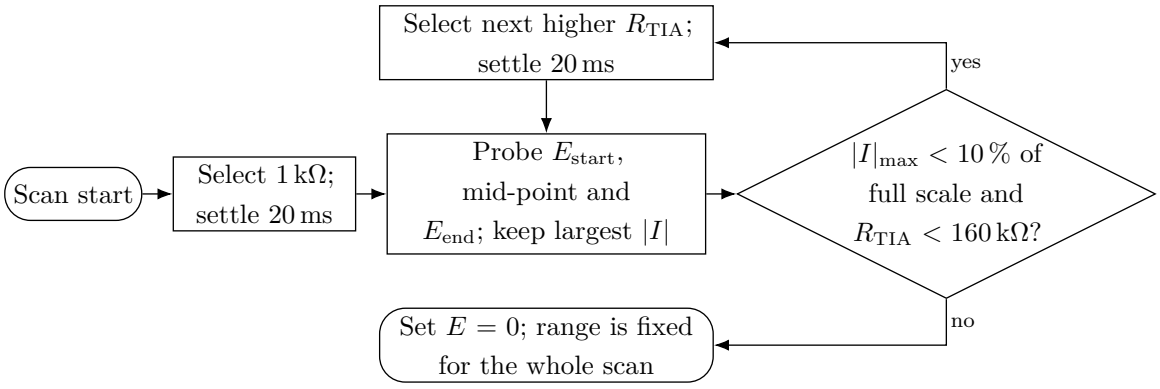
The AD5941 is calibrated against an external resistor connected in place of the cell. The gain and offset constants are compiled into the firmware and are listed as additional information in Appendix F.

5.5.2 Automatic Range Selection

Before a scan the firmware selects the range with the procedure in Figure 5.4. It starts on the 1 kΩ range, which cannot be overloaded by any current the cell is expected to produce, and measures the current at the start, middle and end potentials of the scan with no pulse applied. If the largest of the three is below 10 % of full scale, the next more sensitive range

is selected and the probe is repeated. Successive ranges differ by a factor of five or two, so a signal just below the threshold occupies at most 50 % of full scale on the range that is finally chosen. The remainder is headroom for the pulse response and the Faradaic peak, which the probe does not see because it applies no pulse.

Figure 5.4: Automatic selection of the TIA range before a scan.



The range is then held for the whole scan. Changing the gain resistor during a scan would place the two samples of a step on different ranges, where the offsets no longer cancel in Equation (5.6), and would add a settling transient. If the window-mean TIA voltage exceeds 80 % of full scale (0.72 V) on either sample, the scan is instead stopped, the points captured before the overload are saved and the result is marked as overloaded. The user can also disable automatic selection and choose a fixed range.

5.6 Summary

The firmware runs as a single loop in which one sweep engine serves both the on-device interface and the web dashboard. The DPV waveform is generated by direct DAC writes and timed in software, and each current sample is the mean of 18 filtered ADC results spanning one mains period, placed at the end of each phase and collected without interruption. The current is measured on a range chosen before the scan and held throughout, with a per-range gain and offset correction of which only the gain affects the differential result. The timing accuracy, noise and calibration that result from these choices are evaluated in Chapter 6.

Chapter 6

Testing and Calibration

This chapter evaluates whether the developed potentiostat matches the performance of a commercial one and, where discrepancies occur, whether they originate from the instrument. To isolate error sources, the validation follows a sequence of five progressive tests. Each stage introduces an additional component of a complete electrochemical measurement and benchmarks it against the best available reference standard, allowing any observed deviation to be traced directly to the newly introduced variable:

1. **Excitation waveform:** The potential steps, pulses and timing output by the AFE (Section 6.1).
2. **Resistors:** The current measurement, compared with a digital multimeter (Section 6.2).
3. **Randles circuits:** Dummy cells made from known components and compared against a commercial or laboratory grade instrument, like the PalmSens4 [34] (Section 6.3).
4. **Salt solution:** A commercial or laboratory grade electrode without a redox reaction, against the PalmSens4 (Section 6.4).
5. **Ferricyanide:** An active redox peak, against the PalmSens4 (Section 6.5).

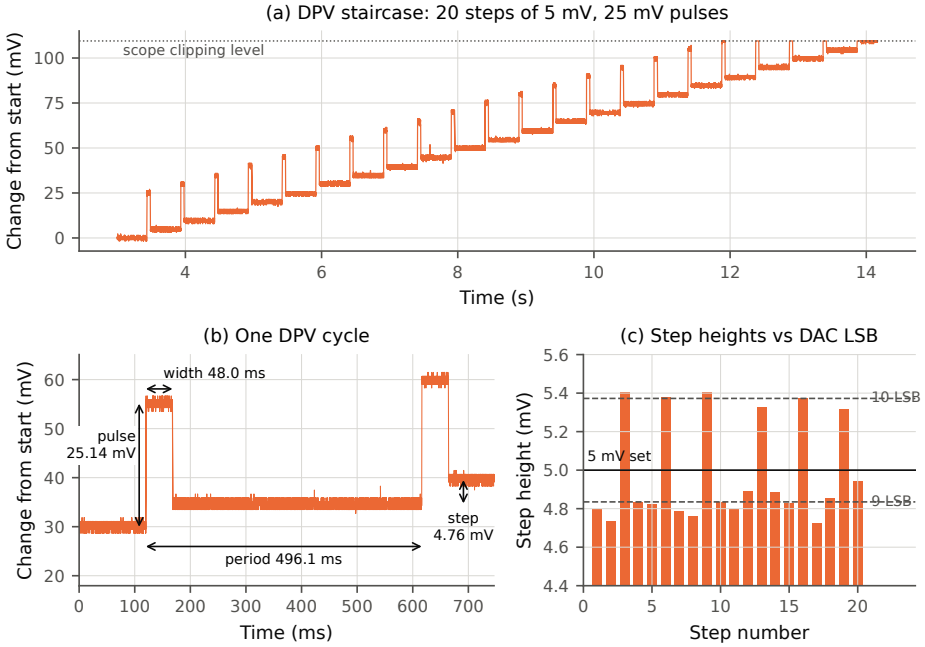
Table 6.1: Measurement conditions.

Electrode	Metrohm DropSens C220AT SPE: 4 mm gold working electrode, gold counter, silver pseudo-reference [62]
Solutions	Blank: 0.1 M KCl. Probe: 5 mM $K_3[Fe(CN)_6]$ in 0.1 M KCl. One drop covering all three electrodes
DPV (Randles)	0.8 V to 1.3 V, step 5 mV pulse 25 mV, 50 ms, interval 0.5 s
DPV (salt, FC)	0.6 V to −0.2 V (KCL) or −0.35 V ($K_3[Fe(CN)_6]$), step 5 mV, pulse 50 mV, 50 ms, interval 0.2 s

6.1 Excitation Waveform

Every later test assumes that the DPV waveform is correctly generated. Hence, the first check is to confirm the output of the device before any current is measured. The excitation was recorded with a Siglent SDS1104X-E oscilloscope while the device ran a DPV scan with a 5 mV step, a 25 mV pulse, a 50 ms pulse width and a 0.5 s interval. The results are shown in Figure 6.1 and Table 6.2.

Figure 6.1: Excitation waveform measured with the oscilloscope. (a) The complete staircase. (b) One cycle with the measured quantities. (c) Height of every step against 9 and 10 LSB of the AD5941 12-bit DAC.



6.1.1 Amplitude

Every step is either 4.82 mV or 5.37 mV and never deviates as illustrated in Figure 6.1c. These are 9 and 10 codes of the 12-bit DAC that sets the cell potential [41]. A 5 mV step is 9.31 LSB, so the firmware rounds each level to the nearest code, and the steps follow the pattern 9, 9, 10 with a mean of 4.98 mV. Because each level is computed from its absolute target rather than by adding steps, the rounding does not accumulate. The pulse is rounded to 47 LSB (25.25 mV) and measures 25.17 mV. The difference between the two step sizes, 0.545 mV, matches the LSB to within the ±3 % gain accuracy of the oscilloscope [63], which limits how precisely the amplitudes can be verified. The applied potential is therefore correct to within LSB. A 0.54 mV resolution is more than enough for DPV, where a peak is normally about 90 mV wide [17].

6.1.2 Timing

The edges settle in less than 0.1 ms, which is negligible against the 50 ms pulse. The pulse width alternates between 48.0 ms and 49.1 ms, and the period between two values about 1 ms apart, consistent with the 1 ms resolution of the firmware timer. Both are shorter than set, the pulse by 1.5 ms on average (3 %) and the period by 3.5 ms (0.7 %).

Timing does not affect the peak potential, which identifies the compound, so these errors change only the peak height. For a diffusion-limited current, which decays as $t^{-1/2}$ [17], sampling 3 % early raises the measured current by about 1.5 %, and the 1.1 ms alternation varies it by about 1 % between points. The device therefore applies the programmed waveform, with amplitudes correct to one LSB and timing within 3 %. With the excitation verified, the next check is the current it measures.

Table 6.2: Measured excitation waveform against the programmed values.

Parameter	Set	Measured	Error
Step height (20 steps)	5 mV	(4.98 ± 0.26) mV	−0.3 %
Pulse height (17 pulses)	25 mV	(25.17 ± 0.08) mV	+0.7 %
Pulse width	50 ms	(48.5 ± 0.6) ms	−3.0 %
Period	500 ms	(496.5 ± 0.6) ms	−0.7 %
Edge time (10–90 %)	–	<0.1 ms	–

6.2 Resistor Calibration

The software converts the high-speed TIA output to a current using the value of the selected gain resistor R_{TIA} [41], so any difference between the nominal and the actual gain of the signal chain appears directly as a current error. Each of the eight ranges was therefore calibrated separately.

A Plusivo DM101 digital multimeter [64] was used to measure a resistor R, which was connected in place of the cell. On its 2 V DC range the multimeter has a resolution of 1 mV and an accuracy of ±0.5 %, and on its 2 kΩ to 200 kΩ resistance ranges an accuracy of ±0.8 %. The value of R was chosen so that ±0.5 V drives a current near the top of its operational range. At ±0.5 V the multimeter was used to measure the voltage V across the resistor, giving the true current $I_{\mathrm{true}} = V/R$, and the firmware reading I_{meas} was recorded. Denoting the two operating points by the subscripts + and −, a linear correction is fitted

through them:

$$I_{\text{true}} = k\, I_{\text{meas}} + c, \qquad k = \frac{I_{\text{true},+} - I_{\text{true},-}}{I_{\text{meas},+} - I_{\text{meas},-}}, \qquad c = I_{\text{true},+} - k\, I_{\text{meas},+}, \tag{6.1}$$

where the gain k corrects the scale and the offset c the zero. The two points lie at opposite ends of the range, so the slope can be estimated accurately. The correction was verified at $+0.25\,\text{V}$, a point not used in the fit.

Before calibration, I_{meas} exceeded I_{true} by 9 % to 12 % on all eight ranges (Table 6.3). An error of the same size and sign on ranges with different resistors points to a cause shared by the conversion chain, such as the assumed ADC reference or PGA gain, rather than to resistor tolerance. After correction the maxiumum error is 0.44 %.

Table 6.3: Two-point calibration of the high-speed TIA ranges, recomputed from the recorded readings. Errors are at the +0.25 V check point.

Range	R (DMM)	Gain k	Offset c (nA)	Error before (%)	Error after (%)
200 Ω	994 Ω	0.9013	160.0	11.36	+0.44
1 kΩ	994 Ω	0.9141	278.7	9.70	+0.40
5 kΩ	4.59 kΩ	0.9089	−69.5	9.99	−0.16
10 kΩ	9.93 kΩ	0.8971	−81.6	11.78	−0.05
20 kΩ	19.9 kΩ	0.9018	−15.8	11.09	+0.06
40 kΩ	38.5 kΩ	0.9129	−9.2	9.72	+0.02
80 kΩ	81.5 kΩ	0.9158	−3.0	9.44	+0.13
160 kΩ	146.7 kΩ	0.9191	1.1	9.15	+0.38

With the gain of every range corrected and checked against an independent reference, the current scale of the device is known to within 0.5 %. This is crucial from an application perspective because the drug concentration is calculated directly from the peak current. The next test uses Randles circuits and adds a capacitive load to the system.

6.3 Randles Circuit Test

The next test adds the capacitance of the cell. A real electrode has a double-layer capacitance, which the DPV sampling of Section 2.1.3 is designed to reject. The Randles test checks this with passive networks of known DC resistance in place of the cell [16]. A total of 7 test cells were used, three cells of the PalmSens4 dummy cell (WE A to C) and four networks built for this project (Cells 1 to 4), each scanned on both instruments. Once the capacitors have charged, the DPV current depends only on the DC path, so the expected value is $\Delta I = \Delta E_{\text{pulse}}/(R_s + R_{ct})$ with $\Delta E_{\text{pulse}} = 25\,\text{mV}$, and zero for the

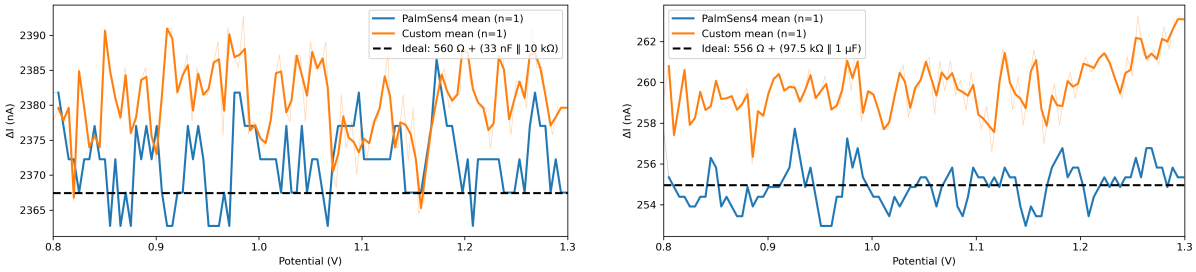
networks with a series capacitor (WE A and Cell 2). The mean ΔI over the scan was compared with this value, and the noise was taken as the standard deviation (Table 6.4).

Table 6.4: Randles circuit test, 0.8 V to 1.3 V

		Expected	Mean ΔI (nA)		Error (%)		Noise (nA)
Cell	**Network**	**(nA)**	**PS4**	**OpenStat**	**PS4**	**OpenStat**	**PS4 / OpenStat**
WE A	560 Ω + (33 nF ∥ diodes)	0	0.00	0.01	–	–	0.10 / 0.07
WE B	10 kΩ	2500.0	2504.4	2511.0	+0.18	+0.44	6.0 / 8.2
WE C	560 Ω + (33 nF ∥ 10 kΩ)	2367.4	2372.0	2380.5	+0.19	+0.55	5.4 / 6.1
Cell 1	93.8 kΩ	266.5	264.9	269.7	−0.62	+1.19	2.2 / 0.8
Cell 2	9.83 kΩ + 470 nF	0.05	0.11	0.21	–	–	0.38 / 0.09
Cell 3	556 Ω + (97.5 kΩ ∥ 1 µF)	255.0	254.9	259.7	−0.02	+1.88	1.0 / 1.4
Cell 4	556 Ω + (923 kΩ ∥ 1 µF)	32.4	33.1	32.4	+2.04	−0.08	0.7 / 1.2

Both instruments measured every network with a DC path to within 2.1 % of the expected value. The OpenStat read slightly high on all of them except Cell 4, by 0.4 % to 0.6 % on the 2.5 µA networks and 1.2 % to 1.9 % on the 260 nA networks. The blocking networks WE A and Cell 2 read below 0.25 nA on both instruments, so the charging current is rejected and there is no DC leakage path. This is the property for which DPV was chosen in this project, and the noise of 0.07 nA to 0.8 nA on the low-current networks is the floor below which the instrument itself cannot resolve a Faradaic current.

Figure 6.2: DPV scans of the dummy cell WE C (left) and Cell 3 (right) from 0.8 V to 1.3 V.



When comparing both potentiostats to the ideal circuit behaviour, the OpenStat matches the expected values and the PalmSens4 to within about 2 %. The next two tests replace the known load with a real electrode. From this point the true value is no longer known, so the PalmSens4 becomes the reference, and its own scan-to-scan variation becomes a variable.

6.4 Salt Solution Test

The fourth test replaces the passive circuit with a real electrode. A drop of 0.1 M KCl on the SPE has a solution resistance and a double-layer capacitance, but no redox reaction in the scan window, which makes it a physical Randles cell with a very large charge-transfer resistance [17]. It should give a smooth background without a peak. The OpenStat is judged against the PalmSens4 measuring the same solution with the same waveform parameters.

Five pairs of salt were compared, Salt 1 to 5, each consisting of one scan on The OpenStat and the PalmSens4. The comparison starts at 0.40 V and finishes at −0.20 V. So that the peak for the ferricyanide test of Section 6.5 would be in the window.

With no analyte present the current measured varies from −185 nA to −210 nA on the PalmSens4 and −192 nA to −200 nA on the OpenStat as shown in Figure 6.3 and Table 6.5. The mean deviation per measured pair ranges from −7.4 % to 4.1 % and the RMS difference from 4.5 nA to 16.8 nA. This is larger than in the Randles test, but is of similar size as the variation of the cell itself. This is shown from scan to scan where the PalmSens4 measurement has a standard deviation of 10.4 nA, and the OpenStat 3.5 nA.

Figure 6.3: Salt solution measured by each potentiostat. The dotted line marks the potential E^* used for the calibration line of Section 6.4.1.

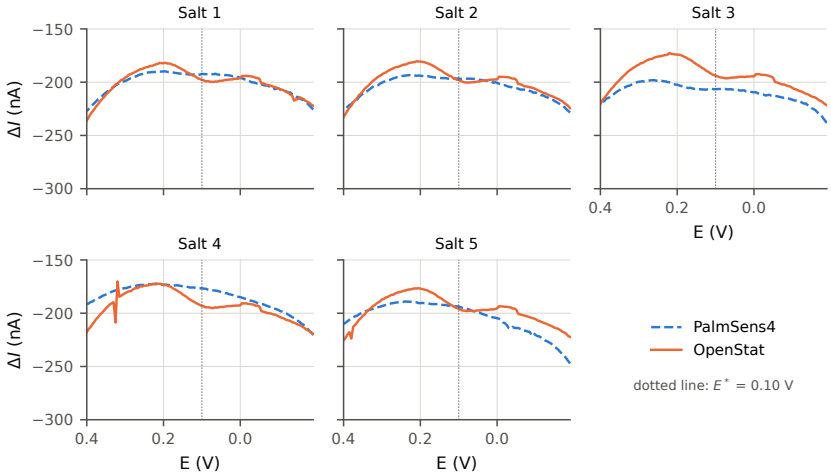


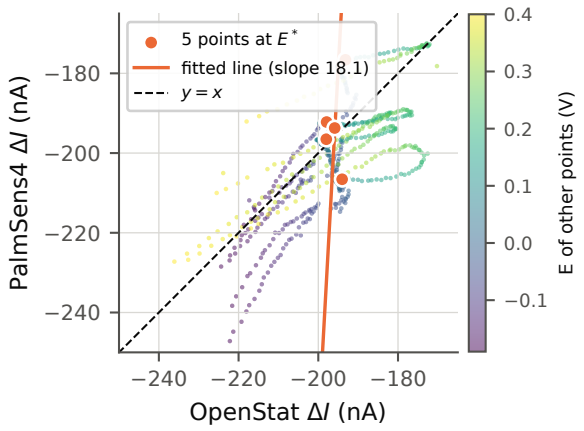
Table 6.5: Salt test, 0.40 V to −0.20 V, all scans on the OpenStat used the 80 kΩ range. r is the correlation coefficient between the PS4 and OpenStat voltammograms of each pair.

	Mean ΔI (nA)		Dev.	RMS	Noise (nA)	
Pair	**PS4**	**OpenStat**	**(%)**	**(nA)**	**PS4 / OpenStat**	r
Salt 1	−200.7	−200.0	−0.3	4.5	0.6 / 0.9	0.94
Salt 2	−204.0	−199.3	−2.3	6.6	0.6 / 0.8	0.93
Salt 3	−209.6	−194.0	−7.4	16.8	0.6 / 0.8	0.90
Salt 4	−184.7	−192.4	+4.1	10.2	0.5 / 3.1	0.86
Salt 5	−204.2	−196.4	−3.8	11.7	0.7 / 1.1	0.80

6.4.1 Calibration Line at a Fixed Potential

An attempt was also made to calibrate the OpenStat against the PalmSens4 using the salt scans. The potential $E^* = 0.10$ V was selected because this is where the ferricyanide peak appears, the OpenStat current is displayed on the x-axis, and the PalmSens4 current on the y-axis. A line $y = ax + b$ was fitted as close as possible to the five points using orthogonal regression and applied at every other potential. The points span only 4.8 nA on the OpenStat axis against 30 nA on the PalmSens4 axis, so the line is almost vertical (slope 18.1) and the points at the other potentials do not follow it (Figure 6.4). It raises the RMS error from 10.8 nA to 225 nA, and is worse than no correction for almost every choice of E^*.

Figure 6.4: Calibration line through the five salt points at $E^* = 0.10$ V, with the points at all other potentials.



Calibration curves are reliable only across their measured range as seen in Section 6.3, and the salt measurements display too little variation to construct one. Section 6.5.3 instead

evaluates the calibration against the ferricyanide peaks.

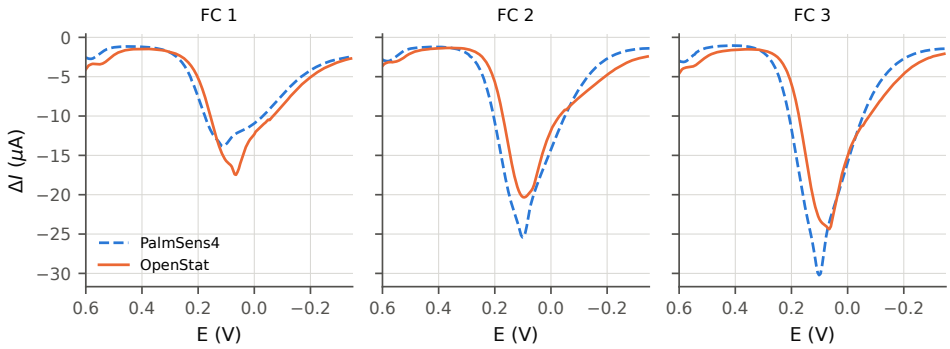
6.5 Ferricyanide Test

The last test adds a redox reaction. Ferricyanide gives a single, well-understood reduction peak, so it tests the potential axis and the current scale together[17].

$$[\mathrm{Fe(CN)_6}]^{3-} + e^- \rightleftharpoons [\mathrm{Fe(CN)_6}]^{4-}, \tag{6.2}$$

All OpenStat scans used the 1 kΩ range. Three pairs were compared, FC 1 to 3, each consisting of one scan on each instrument on the same drop as seen in Figure 6.5.

Figure 6.5: Ferricyanide DPV scans on the same drop, OpenStat run against PalmSens4 run.



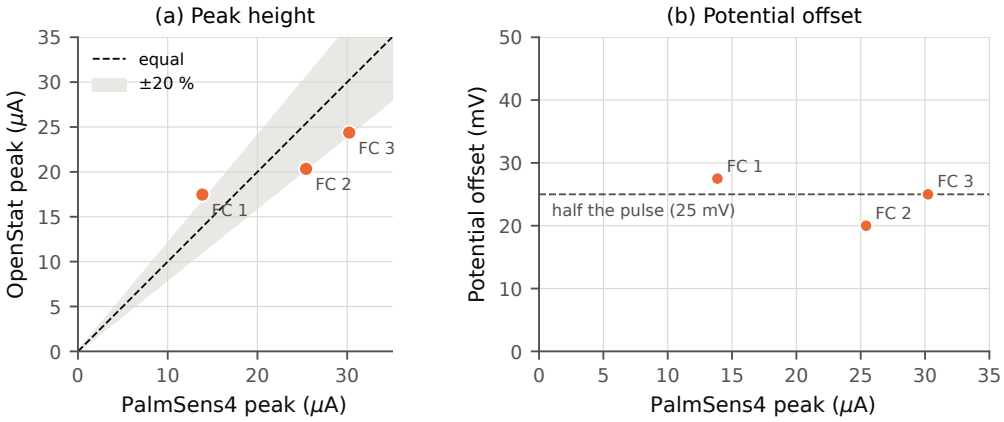
The OpenStat produces a peak similar in shape to the PalmSens4 shown in its correlation $r \geq 0.956$ for all three pairs. However there are two differences visible, the peak heights differ by up to ±26 % and the OpenStat peak lies at a more negative potential (shifted to the right).

6.5.1 Peak Height

The peak heights and potential offsets of the three pairs are plotted in Figure 6.6.

The OpenStat overestimates peak height by 26 % on FC 1 and underestimates it by approximately 20 % on FC 2 and 3, indicating bidirectional deviation rather than regular bias. Crucially, this discrepancy between each devices repeatability must be evaluated. Consecutive PalmSens4 scans of the same droplet were taken to verify this, and they vary by 8 % to 52 %, while consecutive OpenStat scans vary by 3 % to 33 %. Because the mean difference between the different instruments is 22 % which falls within the repeat-measurement variability of the reference device itself, the OpenStat records peak heights very similarly to the PalmSens4 within experimental resolution.

Figure 6.6: The three ferricyanide pairs. (a) Peak height of the OpenStat against the PalmSens4. (b) Potential offset of the OpenStat curve against the peak current.



6.5.2 Peak Potential

The potential offset was found by sliding the OpenStat curve of each pair along the potential axis until it matched the PalmSens4 curve most closely. The offset ranges from 20 mV to 27.5 mV, where 24 mV is the average. By shifiting the OpenStat curve reduces the difference between the curves by up to a third. All the OpenStat ferricyanide scans used the 1 kΩ range, therefore the offset could potentially come from the specific range or its calibration. The calibration cannot cause it, as it only rescales the current. A range-dependent cause would act through a voltage drop proportional to the current, so the offset would grow with the peak, instead it stays the same while the peak more than doubles from FC 1 to FC 3 (Figure 6.6b). Since a peak is about 90 mV wide, an offset of this size must be corrected in firmware before peak potentials are compared with published values. As future work, the offset should also be measured with solutions that fall on other ranges.

6.5.3 Effect of a Calibration Line

In Section 6.5.3 the possibility of making the OpenStat match the PalmSens4 more closely was explored by applying a calibration line fitted to the salt scans. Applied to the three pairs of Figure 6.5, the salt line of Section 6.4.1 multiplies every peak by about 18 (its slope), and the gain-only version changes the mean peak error by less than one percentage point (21.8 % to 22.0 %). Even a line fitted to the ferricyanide drops themselves, which span 6 µA to 30 µA. No amplitude calibration therefore improves the ferricyanide results, the only correction the data supports is the potential offset.

6.5.4 Sources of Deviation

The remaining differences between measurements come from the solution and the testing environment as both instruments see a similar drift. During testing because the two instruments never measure the same drop at the same moment, this drift enters in every comparison, some likely causes are:

- **Adsorption.** Gold is not inert in ferricyanide, when cyanide is released from the complex adsorbs on the gold, it is slowly etched. This results in the active area and electron-transfer rate to change from scan to scan [65, 66]. This could explain the low and varying peak currents and the movement of the peak potential.
- **Light sensitivity.** Ferricyanide decomposes when exposed to light, it breaks down into ferrocyanide and releases cyanide [67]. During every scan the drop on the electrode is exposed to room light, which lowers its ferricyanide concentration and adds cyanide that attacks the gold.
- **Evaporation.** A small drop loses water within minutes, which raises its concentration and explains why repeat scans on one drop grow in peak height rather than shrink.

The comparison would improve by using a fresh electrode and drop for each pair, shielding the drop from light, and starting each scan at a fixed time after the drop is placed.

6.6 Summary

Table 6.6 summarises the five tests and their open points.

Table 6.6: Summary of the tests.

Test	Reference	Result	Open point
Waveform	Oscilloscope	Steps and pulses within one DAC code (0.54 mV), timing within 3 %	Compensate the short pulse and period in firmware
Resistors	DMM	All ranges within 0.44 % at an independent check point (9 % to 12 % before)	Check linearity with more check points per range
Randles	Ideal values, PS4	Within 2.1 %, charging current rejected	Offset of 5 nA to 9 nA, repeat with five scans each
Salt	PS4	Background within ±8 %, inside the scan-to-scan variation, the salt line cannot calibrate	Match the start-of-scan hold to the PalmSens4
Ferricyanide	PS4	Peak heights within the reference's own repeatability, potential offset of about 25 mV	Trace the potential offset in firmware, repeat on the 10 kΩ and 80 kΩ ranges

The bench tests, which rely on isolated electronics show that the instrument itself is accurate. The applied waveform is within one DAC code (0.54 mV) of its set value.

Additionaly, every current range is within 0.44 % after calibration, and the Randles dummy cell is reproduced to within 2.1 % of its ideal values. The salt and ferricyanide tests go a step further. They were added as an extra check that the device performs adequetly in real electrochemistry applications. The results of these tests depend on the electrode, the solution and the scan-to-scan variation of the reference as much as on the potentiostat. The deviations in these tests therefore do not reflect the performance of the electronics. Taking these conditions into account, the device matches the PalmSens4 to within the reference's own repeatability, apart from a potential offset of about 25 mV that can be corrected in firmware.

Chapter 7

Environmental and Sustainability Analysis

An instrument built for environmental monitoring should itself leave a small footprint, so this chapter considers both the problem the OpenStat addresses and its impact over its life.

Antiretrovirals such as efavirenz pass largely untreated through South African wastewater treatment works and reach surface water at up to 102 µg L^{-1} [22], far above the ng L^{-1} levels shown to harm fish [68]. Monitoring is limited by the cost of laboratory analysis, which a low-cost portable instrument helps to lower. The components for the prototype cost about R2 230 including VAT (Appendix L).

The largest savings were made in the hardware itself. The AD5941 integrates all the elements discussed in Section 2.2 into one millimeter-scale package [41], keeping the board small and the part count low. The SPEs, the only consumables, were cleaned using ethanol and reused throughout testing. Table 7.1 lists the remaining measures.

Table 7.1: Further life-cycle measures.

Stage	Design response
Energy	0.89 W measured load (Eq. (4.4)) which equates to ≈ 1500 scans per charge
Obsolescence	Web interface and display which are open-source and can be updated
End of life	Recycling under the EPR Regulations [69]

A full charge draws about 29 W h from a grid emitting 0.960 kg CO_2e/kW h [70], roughly 28 g CO_2e per charge or 0.02 g per scan, so operating emissions are negligible.

The main remaining improvement is in the power chain. Boosting the battery to 5 V and then dropping to 3.3 V discards about a third of the stored energy (Eq. (**??**)). By using a 3.3 V regulator the run time would extend by roughly 50 % with the same cell. Safety, cost and societal considerations are discussed in Appendix **??**.

Chapter 8

Conclusion and Future Work

8.1 Conclusion

This project set out to design an open-source, wireless potentiostat for the DPV detection of antiretroviral drugs in water. The OpenStat meets this aim: it combines the AD5941 analog front-end with an ESP32-S3, runs from a battery, and can be operated either from its own display, buttons and rotary encoder or through a web interface over Wi-Fi. The tests of Chapter 6 showed that the waveform is correct to within one DAC code and that, after calibration, all eight current ranges read within 0.44 %. On dummy cells the device agreed with the expected values to within 2.1 %, and on salt and ferricyanide solutions it matched the PalmSens4 to within the reference instrument's own scan-to-scan repeatability. The only systematic difference is a potential offset of about 25 mV.

8.2 Future Work

On the hardware side, the 200 Ω range cannot be used above about 750 µA because the potentiostat amplifier runs in its normal power mode (Section 6.2). Adding a high-power mode for measurements above this current would make the full range usable. The TP4056 charger has no power-path management, so the device cannot safely run while it charges; a charger with power-path control, would solve this issue.

In the firmware, the OpenStat currently returns the voltammogram and leaves its interpretation to the user. Automatic baseline correction and peak detection would let it report peak height and potential directly, and, combined with standard addition on the same electrode, a concentration. Beyond this, a machine-learning model trained on voltammograms, removed from the scope of this project, could identify which compounds are present where their peaks overlap, rather than only recording the curve. The 25 mV potential offset (Section 6.5.2) should be traced in the firmware and confirmed on the 10 kΩ and 80 kΩ ranges. The device should then be tested with oseltamivir and efavirenz, with preconcentration or a modified electrode to reach nanomolar levels.

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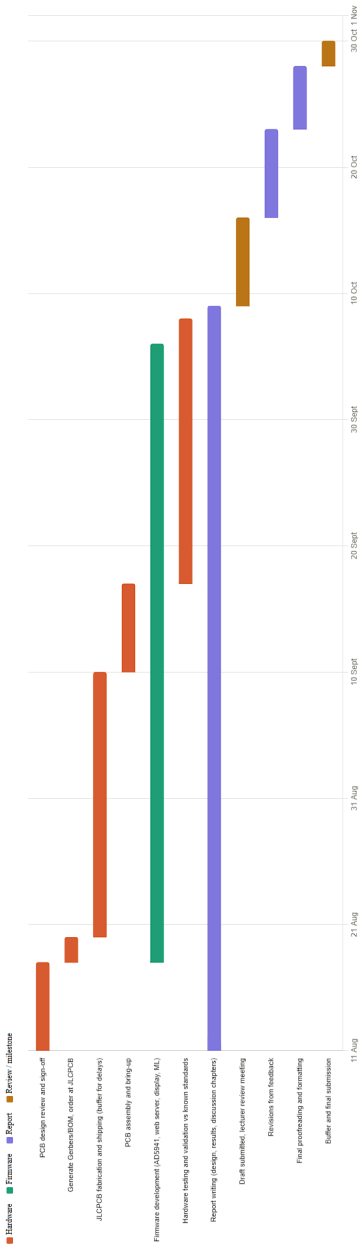
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Appendix A

Project planning schedule

Figure A.1: The decidend Gantt chart for the project planning schedule.



Appendix B

Derivations for the System Model

This appendix gives the derivations behind the results used in Chapter 3.

B.1 Capacitive Current Rejection in DPV

After a potential step, the total current at the electrode is the sum of the Faradaic and capacitive components,

$$I_{\text{total}}(t) = I_f(t) + I_c(t), \tag{B.1}$$

with I_c and I_f given by Equations (2.2) and (2.3). Expanding the two DPV samples into these components gives

$$\Delta I = I_{\text{end}} - I_{\text{before}} = (I_{f,\text{end}} + I_{c,\text{end}}) - (I_{f,\text{before}} + I_{c,\text{before}})\,. \tag{B.2}$$

I_{before} is sampled after the electrode has been held at the baseline potential for long enough that the previous capacitive transient has decayed, so $I_{c,\text{before}} \approx 0$. I_{end} is sampled at the end of a pulse that is long compared with $R_s C_{dl}$, so the transient caused by the pulse has also decayed and $I_{c,\text{end}} \approx 0$. Equation (B.2) therefore reduces to Equation (2.4), leaving only the Faradaic signal.

Because ΔI is the change in Faradaic current caused by a small potential change ΔE, it approximates the derivative of the current–potential curve:

$$\Delta I \approx \frac{\mathrm{d}I_f}{\mathrm{d}E}\,\Delta E. \tag{B.3}$$

For a reversible system governed by the Nernst equation, the current–potential curve is sigmoidal, and its derivative is a peak centred close to the half-wave potential $E_{1/2}$, offset from it by half the pulse amplitude [17, 71].

B.2 Negative-Feedback Stability

Both the control loop and the TIA loop have the structure of Figure B.1, where $A(s)$ is the amplifier and $\beta(s)$ is the feedback network.

Figure B.1: Control loop block diagram for the analog front-end.

The closed-loop transfer function is [42]

$$T(s) \equiv \frac{e_o(s)}{E_{\text{in}}(s)} = \frac{A(s)}{1 + A(s)\beta(s)}, \tag{B.4}$$

and the loop gain is

$$L(s) \equiv A(s)\beta(s). \tag{B.5}$$

The poles of $T(s)$, which govern the transient response, are the roots of the characteristic equation $1 + L(s) = 0$. At the crossover frequency ω_c, where $|L(j\omega_c)| = 1$, the phase margin is

$$\phi_{\text{PM}} = 180^\circ + \angle L(j\omega_c). \tag{B.6}$$

If $\phi_{\text{PM}} \leq 0^\circ$, the closed-loop poles lie in the right half of the s-plane and the system oscillates [42].

B.3 Control-Loop Feedback Factor

In the reduced cell of Figure **??**, R_u and C_{dl} form a series path from the control amplifier output $e_o(s)$ to the virtual ground at the WE. The current through this path is

$$I(s) = \frac{e_o(s)}{R_u + \dfrac{1}{sC_{dl}}}, \tag{B.7}$$

and the RE senses the voltage across the capacitive branch,

$$E_{\text{ref}}(s) = I(s) \cdot \frac{1}{sC_{dl}}. \tag{B.8}$$

Combining the two gives the feedback factor of Equation (**??**):

$$\beta(s) \equiv \frac{E_{\text{ref}}(s)}{e_o(s)} = \frac{1}{1 + sR_uC_{dl}}. \tag{B.9}$$

This single-pole cell model is the one used in the classical potentiostat instrumentation literature to analyse control-loop bandwidth and IR-compensation limits [17, 72, 73].

With $A(s) = \omega_{t,\text{ctrl}}/s$, the magnitude of the loop gain in Equation (**??**) is

$$|L_{\text{ctrl}}(j\omega)| = \frac{\omega_{t,\text{ctrl}}}{\omega\sqrt{1 + \omega^2\tau_{\text{cell}}^2}}. \tag{B.10}$$

For $\omega\tau_{\text{cell}} \gg 1$ this becomes $\omega_{t,\text{ctrl}}/(\omega^2\tau_{\text{cell}})$, and setting it to unity gives Equation (**??**). The phase of the loop gain is $-90^\circ - \arctan(\omega\tau_{\text{cell}})$, so the phase margin is

$$\phi_{\text{PM,ctrl}} = 90^\circ - \arctan(\omega_{c,\text{ctrl}}\tau_{\text{cell}}) \approx 90^\circ - \arctan\sqrt{\omega_{t,\text{ctrl}}\,\tau_{\text{cell}}}, \tag{B.11}$$

which decreases as τ_{cell} increases.

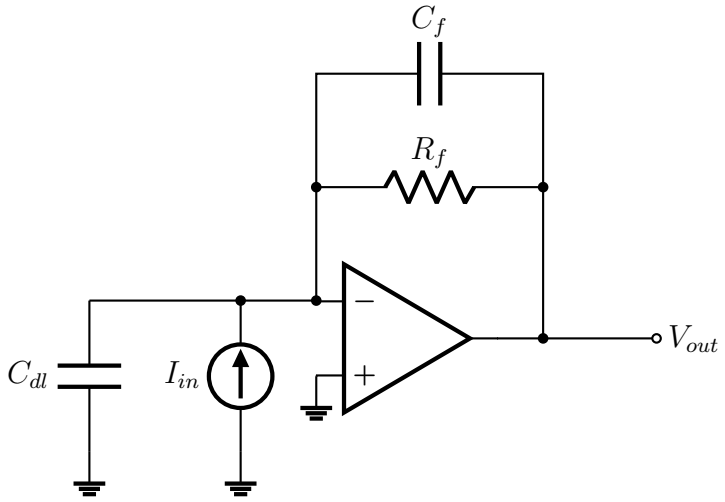
The RE buffer is a voltage follower, so $\beta = 1$ and

$$T_{\text{buf}}(s) = \frac{A_{\text{buf}}(s)}{1 + A_{\text{buf}}(s)} \approx 1 \quad \text{for } |A_{\text{buf}}(s)| \gg 1, \tag{B.12}$$

so it contributes no pole over the frequency range of interest [74].

B.4 TIA Transfer Function and Compensation

Figure B.2: Analog front-end transimpedance amplifier circuit.



Applying Kirchhoff's current law at the inverting node of the TIA in Figure B.2, without C_f, the cell current I_{in} divides between C_{dl} and R_f:

$$I_{\text{in}} = V_n\, sC_{dl} + \frac{V_n - V_{\text{out}}}{R_f}. \tag{B.13}$$

With a finite open-loop gain, the inverting node is not exactly at zero:

$$V_n = -\frac{V_{\text{out}}}{A(s)}. \tag{B.14}$$

Substituting Equation (B.14) into Equation (L.1) gives the closed-loop transimpedance

$$Z_T(s) \equiv \frac{V_{\text{out}}}{I_{\text{in}}} = -\frac{R_f}{1 + \dfrac{1 + sR_fC_{dl}}{A(s)}} = -\frac{R_f}{1 + \dfrac{NG(s)}{A(s)}}, \tag{B.15}$$

which identifies the noise gain $NG(s) = 1 + sR_fC_{dl}$ of Equation (**??**) and the loop gain $A(s)/NG(s)$ of Equation (**??**) [28, 29]. Setting $|L_{\text{TIA}}(j\omega)| = 1$ for $\omega R_fC_{dl} \gg 1$ gives Equation (**??**) in the same way as for the control loop.

With C_f in parallel with R_f, the feedback impedance becomes $Z_f = R_f/(1 + sR_fC_f)$ and the noise gain becomes

$$NG(s) = 1 + sC_{dl}Z_f = \frac{1 + sR_f\,(C_{dl} + C_f)}{1 + sR_fC_f}. \tag{B.16}$$

The zero lies at $1/[2\pi R_f(C_{dl} + C_f)]$ and the new pole at $f_p = 1/(2\pi R_fC_f)$, which is above the zero since $C_f < C_{dl} + C_f$. Placing the pole at the geometric mean of the zero and the GBW, with $C_f \ll C_{dl}$ so that $f_z \approx 1/(2\pi R_fC_{dl})$,

$$\frac{1}{2\pi R_fC_f} = \sqrt{\frac{\text{GBW}}{2\pi R_fC_{dl}}}, \tag{B.17}$$

and solving for C_f gives Equation (**??**).

Appendix C

Display and SD Card Specifications

Table C.1: Key specifications of the DFRobot DFR0928 display [2].

Parameter	**Value**
Driver	ST7735S
Resolution	128×160 pixels
Interface	4-wire SPI
Supply current	33 mA at 3.3 V
Storage	SD card slot

Appendix D

Power Management Subsystem Specifications

Table D.1: Main specifications of the power subsystem components.

TP4056 [57]	**DW01A** [75]	**S9V11E2F5** [58]
Input supply voltage (V_{CC}): 4.0 V to 8.0 V	Overcharge detection: approx. 4.28 V to 4.30 V	Output voltage: fixed 5 V (+5 % / −3 %)
Preset charge voltage: 4.2 V with 1.5 % accuracy	Overdischarge detection: approx. 2.40 V	Input voltage range: 2 V to 16 V
Charge current: programmable up to 1000 mA	Overcurrent protection	Start-up voltage: 3 V; start-up current limit: 700 mA
Trickle charge threshold: 2.9 V	Low current consumption (1.5 µA)	Typical maximum output current: up to 1.7 A
Operating temperature range: −40 °C to 85 °C		Efficiency: typically 85 % to 95 %

Appendix E

Scan Flow and Timing Diagrams

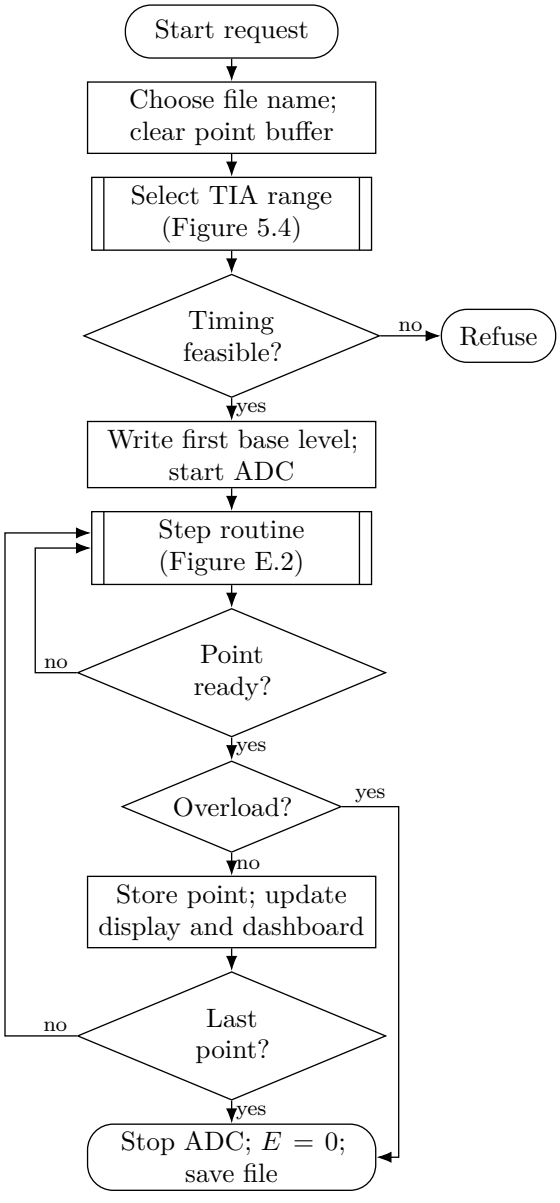


Figure E.1: Flow of one DPV scan.

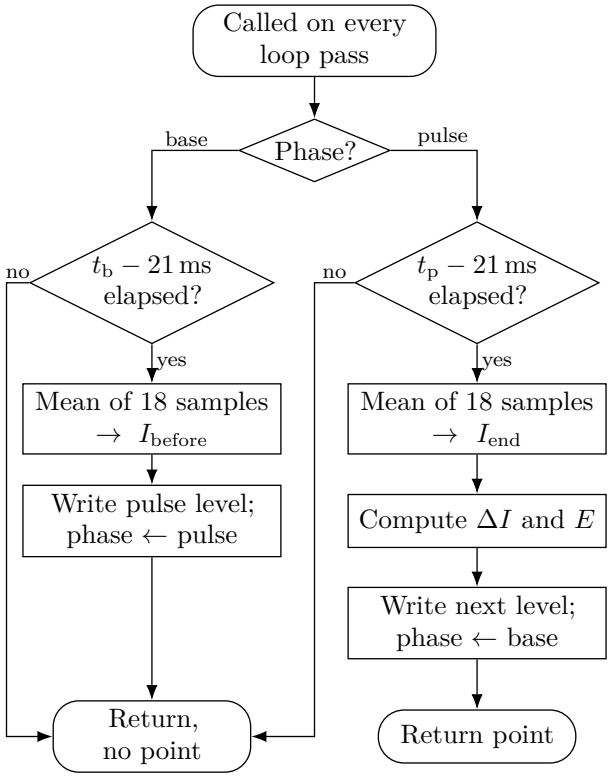


Figure E.2: Step routine. $t_b = T - t_p$ is the base dwell.

Appendix F

RTIA Calibration Constants and Resistors

Table F.1: Current ranges of the high-speed TIA, the calibration constants held in the firmware and the resistor each range was calibrated against.

Code	R_{TIA}	**Full scale**	**Resolution**	G_k	O_k (nA)	**Cal. resistor**
0	200 Ω	±4.5 mA	187 nA	0.9013	+160.04	994 Ω
1	1 kΩ	±900 µA	37.3 nA	0.9141	+278.66	994 Ω
2	5 kΩ	±180 µA	7.47 nA	0.9089	−69.54	4.59 kΩ
3	10 kΩ	±90 µA	3.73 nA	0.8971	−81.57	9.93 kΩ
4	20 kΩ	±45 µA	1.87 nA	0.9018	−15.8	19.9 kΩ
5	40 kΩ	±22.5 µA	0.93 nA	0.9129	−9.159	38.5 kΩ
6	80 kΩ	±11.3 µA	0.47 nA	0.9158	−3.014	81.5 kΩ
7	160 kΩ	±5.6 µA	0.23 nA	0.9191	+1.005	146.7 kΩ

Appendix G

Webserver Dashboard

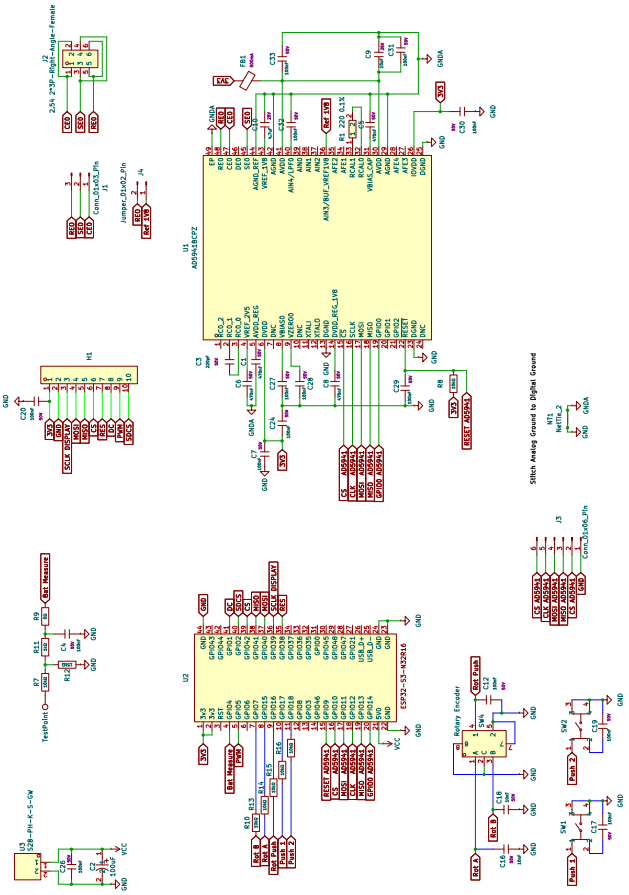
Figure G.1: The webserver dashboard for the potentiostat.



Appendix H

KiCad Schematic

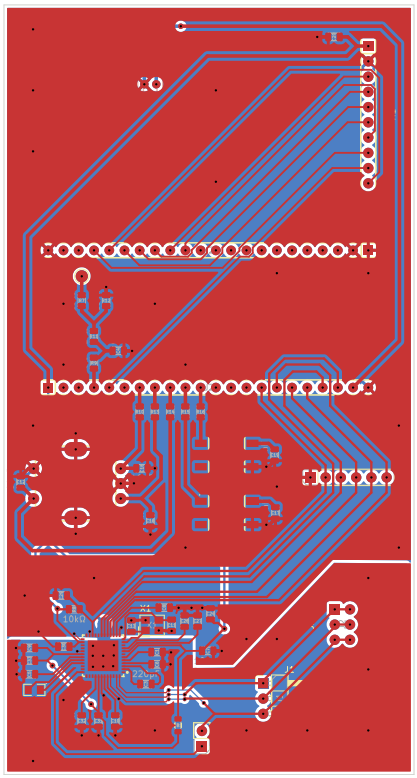
Figure H.1: The KiCad schematic for the potentiostat.



Appendix I

KiCad PCB

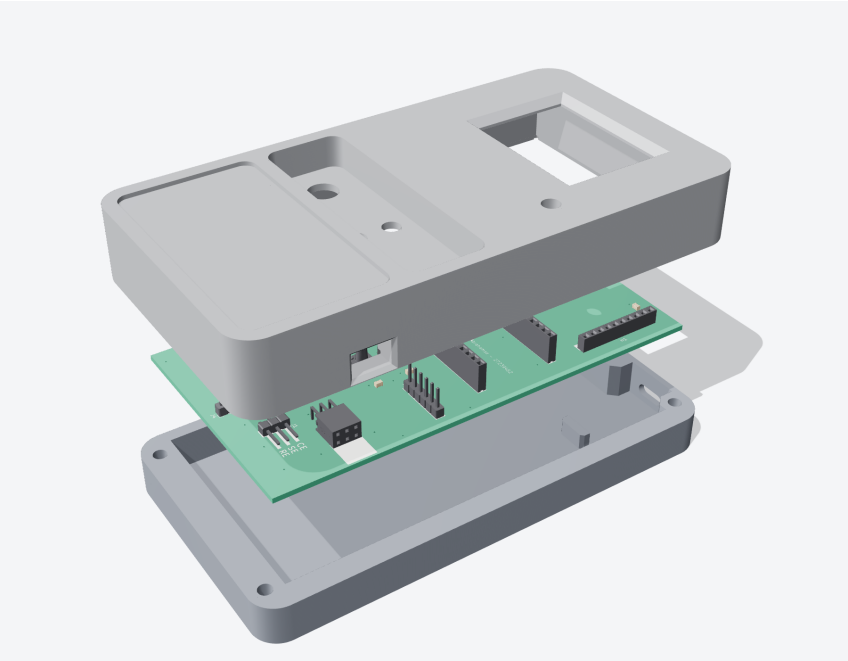
Figure I.1: The KiCad PCB for the potentiostat.



Appendix J

Case Render

Figure J.1: The case render for the potentiostat.



Appendix K

AD5941 Capacitor Selection

This appendix gives the full reasoning behind the capacitor values summarised in Table 4.3 of Section 4.4.1.

K.1 Supply Decoupling: 100 nF and 10 µF

Each supply pin has a local 100 nF ceramic capacitor, and AVDD has an additional 10 µF bulk capacitor. The MT-101 tutorial prescribes two tiers of capacitors: a large capacitor, typically 10 µF to 100 µF, within about 5 cm of the IC, and small ceramics, typically 0.01 µF to 0.1 µF, as close to the power pins as possible, optionally with a ferrite bead in series [45]. Analog Devices applies the same principle in its precision converter datasheets; for the AD7714, for example, a 10 µF capacitor is used in parallel with 0.1 µF on each analog supply [76].

Target impedance. The worst-case AVDD current step is the sum of all the subsystems of the AD5941 switching on at once [41, p. 14], which is approximately 10 mA. The 1.82 V reference has an AC PSRR of 60 dB [41, p. 11], which corresponds to supply noise being attenuated by a factor of about 1000. A 10 mV supply disturbance therefore appears as 10 µV on the reference, which is a factor of 5.5 below 1 LSB $= 1.82\,\mathrm{V}/2^{15} = 55.5\,\mu\mathrm{V}$ [41, p. 7]. Taking 10 mV as a conservative limit on the supply disturbance, the target impedance is

$$Z_t = \frac{\Delta V}{\Delta I} = \frac{10\,\mathrm{mV}}{10\,\mathrm{mA}} = 1\,\Omega. \tag{K.1}$$

10 µF. The impedance of a capacitor falls as frequency rises, so a given capacitor holds the rail below Z_t only above a certain frequency. Setting the capacitor impedance equal to Z_t and solving for frequency gives that lower limit:

$$|Z_C| = \frac{1}{2\pi f C} = Z_t \quad \Rightarrow \quad f_{\min} = \frac{1}{2\pi C Z_t} = \frac{1}{2\pi\,(10\,\mu\mathrm{F})(1\,\Omega)} = 15.9\,\mathrm{kHz}. \tag{K.2}$$

Above 15.9 kHz the capacitor can supply the sudden current demands of the chip quickly enough that the AVDD voltage barely moves. Below 15.9 kHz the 3.3 V regulator holds

the rail stable.

100 nF. Rapid switching inside the AD5941 draws short current bursts from the supply. The 10 µF capacitor cannot deliver bursts this fast. A small capacitor placed directly at each supply pin has far less inductance in its path and can deliver them, covering the frequencies from a few megahertz to beyond 100 MHz. A value of 100 nF is recommended for this role in standard decoupling guidance [45, 51, 77], and the calculations below show that it is adequate for this design.

Every capacitor resonates with its own equivalent series inductance (ESL) at its self-resonant frequency, above which it behaves as an inductor. With a typical ESL of about 1 nH for the 0805 package used on this board [**TODO: reference for 0805 ESL**]:

$$f_{\mathrm{SRF}} = \frac{1}{2\pi\sqrt{L_{\mathrm{ESL}}\,C}}: \qquad f_{\mathrm{SRF},\,10\,\mu\mathrm{F}} = 1.6\,\mathrm{MHz}, \qquad f_{\mathrm{SRF},\,100\,\mathrm{nF}} = 15.9\,\mathrm{MHz}. \tag{K.3}$$

The 10 µF capacitor is most effective up to around 1.6 MHz, which makes it the right part for the slower range, roughly 16 kHz to a few megahertz. The 100 nF capacitor keeps improving up to around 15.9 MHz. Above that its impedance is set by the ESL alone,

$$|Z| \approx 2\pi f L_{\mathrm{ESL}} = 2\pi\,(100\,\mathrm{MHz})(1\,\mathrm{nH}) = 0.63\,\Omega, \tag{K.4}$$

which is still below the 1 Ω target, so it covers the fast range adequately.

K.2 Internal Nodes: 470 nF and 4.7 µF

These capacitors connect to nodes inside the AD5941, specifically its voltage references and internal regulators. Their correct values depend on internal circuit details that Analog Devices does not publish, so they cannot be derived independently and the datasheet values are used directly.

470 nF. The datasheet specifies 470 nF for three pins, namely `VREF_2V5`, `VBIAS_CAP` and `DVDD_REG_1V8` [41, pp. 11, 20–21], but does not explain how this value was chosen. Capacitors on such nodes are normally sized to suit the stability and noise requirements of the on-chip reference buffers and regulators [48], and those requirements cannot be assessed without internal design data. The datasheet value was therefore adopted as specified.

The datasheet gives no value for `AVDD_REG` [41, p. 20]. The 470 nF specified for the other internal regulator output, `DVDD_REG_1V8`, was used as the closest available guidance.

4.7 µF. The datasheet pin description recommends 4.7 µF on `VREF_1V82`, the 1.82 V reference of the ADC [41, p. 19], again without stating the reasoning. Its specification

table, however, lists 0.47 µF as the test condition for the same pin [41, p. 11], so the datasheet is not internally consistent. The 4.7 µF recommendation was followed as the more conservative of the two. The ADC draws a small pulse of charge from this reference during every conversion, and a larger capacitor holds more charge, so the reference voltage dips less with each pulse.

Appendix L

Bill of Materials

Table L.1 lists the components bought for the prototype. Prices are the suppliers' listed prices in rand excluding VAT, for the quantity ordered; items sold in packs or with a minimum order are costed at the full pack. Shipping and import duties are excluded. Suppliers are abbreviated as MR (MicroRobotics), DK (DigiKey), CM (Communica), RS (RS Components) and PW (PCBWay). Items marked † are estimates: the resistors and capacitors marked Kit came from laboratory component kits and are costed at Communica's single-unit price for the same value (the 12 pF at the nearest listed value, 15 pF), and the PCB is costed from a PCBWay online quote of US$30.89 for five boards at R16.59/US$ (25 October 2026).

Table L.1: Bill of materials for the OpenStat prototype.

Description	Part no.	Supplier	Qty	Unit (R)	Total (R)
Microcontroller					
ESP32-S3 development board, 32 MB flash, 16 MB PSRAM	N32R16V-M	MR	1	336.00	336.00
Display and storage					
1.8-inch 128 × 160 TFT display with microSD slot	DFR0928	MR	1	168.00	168.00
32 GB microSD card	MSC100-32GB	MR	1	148.00	148.00
Power					
Li-Po cell, 3.7 V, 5000 mA h	955565	MR	1	138.00	138.00
Li-Po charger module, 1 A, USB-C (TP4056)	PK4056	MR	1	16.00	16.00
Step-up/step-down regulator, 5 V	Pololu S9V11E2F5	MR	1	80.00	80.00
Electrolytic capacitor, 100 µF 25 V (pack of 10)	CAP-100UF-25V	MR	1	8.00	8.00
Analog front end					
Ferrite bead 0805, 600 Ω at 100 MHz	74279204	RS	10	3.98	39.84
Crystal, 16 MHz, ±10 ppm	X1E000021011912	RS	5	33.63	168.16
Thin-film resistor 0805, 220 Ω, 0.1 %	ERA6VEB2200V	RS	10	7.62	76.20
Connectors					
JST-PH 2-pin cable, 20 cm (pack of 4)	JST-PH-2P-20CM	MR	1	10.00	10.00

Table L.1 continued

Description	Part no.	Supplier	Qty	Unit (R)	Total (R)
Edge connector, dual-row female, 6-way	7-5530843-7	DK	2	49.85	99.70
DIL socket, 6-way, right-angle	727060	CM	2	3.15	6.30
SIL header, 40-way	709401	CM	1	6.25	6.25
Female header, 40-way (pack of 10)	H40P-FEM-254	MR	1	18.00	18.00
SIL header, 20-way (pack of 10)	HM20P-254-BLK	MR	1	10.00	10.00
User input					
Rotary encoder EC11, vertical	EC11-VS-15	MR	2	10.00	20.00
Tactile switch, 6 mm × 6 mm SMD	DTSM63R	CM	4	3.70	14.80
Passives					
SMD electrolytic capacitor, 100 µF 6.3 V	100UF 6,3VES	CM	4	1.35	5.40
Resistor 0805, 0 Ω (minimum order 100)	CHR0805 0R	CM	100	0.14	14.00
Capacitor 0805, 470 nF†	CHC0805	Kit	4	0.60	2.40
Capacitor 0805, 4.7 µF†	CHC0805	Kit	1	0.56	0.56
Capacitor 0805, 12 pF C0G†	CHC0805	Kit	2	0.35	0.70
Capacitor 0805, 100 nF†	CHC0805	Kit	15	0.94	14.10
Capacitor 0805, 10 nF†	CHC0805	Kit	2	0.45	0.90
Capacitor 0805, 220 pF C0G†	CHC0805	Kit	1	0.20	0.20
Capacitor 0805, 10 µF†	CHC0805	Kit	1	1.04	1.04
Resistor 0805, 10 kΩ, 1 %†	CHR0805	Kit	7	0.12	0.84
Resistor 0805, 1 kΩ, 1 %†	CHR0805	Kit	1	0.12	0.12
Resistor 0805, 15 kΩ, 1 %†	CHR0805	Kit	1	0.12	0.12
Prototyping					
Veroboard, 50 mm × 70 mm (pack of 4)	VERO5070	MR	1	20.00	20.00
PCB					
Two-layer FR-4 PCB, 150 mm × 50 mm (5 boards)†	PCBWay	PW	5	102.48	512.41
Subtotal (excl. VAT)					1 936.04
VAT (15 %)					290.41
Total (incl. VAT)					**2 226.45**

L.1 TIA Noise Gain with a Series Cell

At the inverting input of the TIA the cell is represented by the impedance $Z_c = R_\Sigma + 1/sC_{dl}$ to the solution node, which the control loop holds at a fixed potential, and the feedback network by $Z_f = R_{\mathrm{TIA}}/(1 + sR_{\mathrm{TIA}}C_f)$. Applying Kirchhoff's current law at the inverting node, with V_n the node voltage and I_{in} the faradaic current,

$$I_{\mathrm{in}} = \frac{V_n}{Z_c} + \frac{V_n - V_{\mathrm{out}}}{Z_f}, \qquad V_n = -\frac{V_{\mathrm{out}}}{A(s)}. \tag{L.1}$$

Eliminating V_n gives the closed-loop transimpedance

$$Z_T(s) \equiv \frac{V_{\text{out}}}{I_{\text{in}}} = -\frac{Z_f}{1 + NG(s)/A(s)}, \qquad NG(s) = 1 + \frac{Z_f}{Z_c}, \tag{L.2}$$

so the loop gain is $A(s)/NG(s)$ [28]. With $C_f = 0$,

$$NG(s) = 1 + \frac{sR_{\text{TIA}}C_{dl}}{1 + sR_\Sigma C_{dl}} = \frac{1 + s\,(R_{\text{TIA}} + R_\Sigma)\,C_{dl}}{1 + sR_\Sigma C_{dl}}, \tag{L.3}$$

which is Equation (3.1). Setting $R_\Sigma = 0$ recovers the photodiode case, $NG = 1 + sR_{\text{TIA}}C_{dl}$, whose noise gain rises without limit.

With $A(s) = 2\pi\,\text{GBW}/s$, the phase of the loop gain is

$$\angle L(jf) = -90^\circ - \arctan\frac{f}{f_z} + \arctan\frac{f}{f_p}. \tag{L.4}$$

For $f_c \gg f_z$ the second term is close to -90°, so $\phi_{\text{PM}} = 180^\circ + \angle L(jf_c) \approx \arctan(f_c/f_p)$, which is Equation (3.2). Above f_p, $|NG| \approx 1 + R_{\text{TIA}}/R_\Sigma$ and $|A| = \text{GBW}/f$, so the crossover is at $f_c = \text{GBW}\,R_\Sigma/(R_{\text{TIA}} + R_\Sigma)$. A margin of 45° requires $f_c \geq f_p$:

$$\frac{\text{GBW}\,R_\Sigma}{R_{\text{TIA}} + R_\Sigma} \geq \frac{1}{2\pi R_\Sigma C_{dl}} \quad\Longrightarrow\quad R_\Sigma^2 \geq \frac{R_{\text{TIA}} + R_\Sigma}{2\pi\,\text{GBW}\,C_{dl}} \approx \frac{R_{\text{TIA}}}{2\pi\,\text{GBW}\,C_{dl}}, \tag{L.5}$$

since $R_\Sigma \ll R_{\text{TIA}}$, which gives Equation (3.3).

With C_f fitted, Z_f falls above $1/(2\pi R_{\text{TIA}}C_f)$, so the noise gain falls from its plateau at $-20\,\text{dB}$ per decade and adds phase lead at the crossover. Because the approximations above lose accuracy when f_c lies near f_p, the margins in Table 3.1 were computed by solving $|A(jf)/NG(jf)| = 1$ numerically with the full expression for $NG(s)$, including C_f.